

Incremental Predictive Value of Bullying Exposure Indicators for Predicting Persistent Sadness or Hopelessness in the 2023 National Youth Risk Behavior Survey

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Abstract

This study examines whether two bullying exposure indicators from the 2023 National Youth Risk Behavior Survey improve prediction of persistent sadness or hopelessness beyond a narrow demographic baseline. The outcome is QN26. The focal predictors are QN24 bullied on school property and QN25 electronically bullied. The analysis uses a fixed held-out validation design, a frozen train-test split, calibration-focused model selection, and research-only claim boundaries. Weighted summaries are used only for descriptive context. Predictive comparisons rely on unweighted held-out evaluation. The final comparison contrasts baseline demographic models with models that add the bullying block and evaluates ROC AUC, PR AUC, Brier score, and calibration. Among the full-feature candidates, histogram gradient boosting with Platt scaling achieved held-out ROC AUC of 0.716, PR AUC of 0.644, Brier score of 0.206, calibration slope of 1.005, and calibration intercept of 0.014. Adding the bullying indicators improved held-out ROC AUC by 0.066, PR AUC by 0.106, and Brier score by 0.019 relative to the non-bullying comparator. The results support the hypothesis that bullying exposure indicators add measurable incremental predictive value in this bounded design, while remaining predictive associations rather than causal or deployment-ready conclusions.

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1. Introduction

1.1 Adolescent Mental Health, Surveillance, and the YRBS Frame

Adolescent mental health remains a central public health concern, and the 2023 Youth Risk Behavior Survey provides a nationally important surveillance frame for studying that concern at population scale. Brener et al. (2024) describe the Youth Risk Behavior Surveillance System as a recurring public health monitoring system designed to track health-related behaviors and outcomes across U.S. high school students through a documented survey design. The Centers for Disease Control and Prevention (2024) provide the companion user guidance that governs how the 2023 public-use files should be interpreted. Those sources matter because this thesis is not operating in a vacuum. It is nested inside a surveillance instrument with stated coding rules, design variables, weighting logic, and clear limits on what secondary analysis can claim.

That surveillance frame does two things for the thesis. First, it establishes why persistent sadness or hopelessness deserves careful attention. The 2023 weighted prevalence of QN26 in the local project outputs is 39.6 percent, which is high enough to justify a focused probability-oriented study rather than a marginal descriptive note. Second, it defines the discipline of the thesis. The YRBS is designed for population surveillance, not for diagnosis, triage, or causal adjudication. The present thesis therefore treats the survey as a public data source for a narrow research-stage prediction problem rather than as a stand-in for clinical or institutional decision making.

The distinction between those two uses is foundational. Weighted descriptive context helps explain why the problem matters at the national level. It does not determine the predictive claim. The predictive claim depends on whether a locked predictor block changes held-out probability quality once the baseline covariate set is already fixed. Throughout the thesis, that distinction keeps the document anchored to the strengths of the data rather than to ambitions the data cannot support.

1.2 Why Persistent Sadness or Hopelessness Is the Correct Thesis Target

The target outcome in this thesis is QN26 persistent sadness or hopelessness. This is the right focal outcome for three reasons. First, it is the outcome locked by the proposal and preserved throughout the study design. Second, it captures a broad distress signal that is central to the public-health and bullying literatures reviewed in Chapter 2. Third, it supports a probability-focused predictive analysis with enough prevalence to make calibration, Brier score, and held-out probability quality scientifically meaningful.

The choice to remain with a single target is methodological discipline, not conservatism for its own sake. Expanding to a larger family of mental health outcomes could increase apparent document size, but it would blur the prespecified protocol and weaken the clarity of the central

comparison. A thesis benefits from scope control when that control sharpens the causal, predictive, and governance boundaries of the work. Here, one locked outcome gives the manuscript a stronger center of gravity than a broader but less coherent outcome set would.

The target also works well with the held-out evaluation philosophy adopted later in the manuscript. Because QN26 is common enough to support probability scoring, the thesis can judge models on more than ranking alone. That matters because the project is not asking only whether bullied students are, on average, at greater observed risk. It is asking whether a bullying block changes the quality of predicted probabilities for this one clearly defined outcome after demographic adjustment.

1.3 Why Bullying Indicators Matter as Predictive Inputs

The focal predictors are QN24 bullied on school property and QN25 electronically bullied. These variables matter in the thesis because they are plausible predictive indicators, not because the thesis is trying to elevate them into stand-alone causal explanations. The literature already shows that bullying victimization is associated with worse mental health outcomes among adolescents. Young et al. (2024) reported that frequent social media use, bullying victimization, persistent sadness or hopelessness, and suicide risk clustered meaningfully in the 2023 YRBS. Ye et al. (2023) showed in a broader meta-analytic context that bullying victimization is associated with depressive symptoms in children and adolescents. Grimland et al. (2025) and Hinduja and Patchin (2025) similarly show that cyberbullying matters for serious adolescent mental health outcomes and should not be collapsed into a trivial extension of offline conflict.

That literature makes a predictive question reasonable. It does not answer it. A predictor can be substantively relevant and still fail to improve held-out prediction once a baseline adjustment set is in place. Conversely, a predictor can improve held-out prediction without settling the causal structure behind the improvement. The thesis lives inside that narrower predictive space. It tests whether school-property bullying and electronic bullying alter the observed probability surface for QN26 beyond a locked demographic baseline.

Treating the bullying indicators as a block rather than as isolated coefficients is also deliberate. The study asks whether the paired exposure signals together add predictive value under a pre-specified comparison rule. That framing is cleaner than a coefficient-by-coefficient reading because it keeps the thesis focused on incremental predictive value of a substantively motivated feature block rather than on over-interpreting one fitted parameter at a time.

1.4 The Methodological Stakes Are Held-Out Probability Quality

The strongest methodological stake in this thesis is not whether one model can post a superficially impressive ROC AUC. It is whether the final thesis can defend a probability-

oriented prediction claim under a design that is tight enough to be credible. That is why calibration, Brier score, held-out evaluation, and lexicographic model selection are treated as first-order components of the research design. Huang et al. (2020) and Silva Filho et al. (2023) argue that predicted class probabilities must be assessed for calibration rather than being judged only by discrimination. Efthimiou et al. (2024) likewise emphasize transparent specification and validation for prediction studies. Those expectations matter even more in a public health setting where the numbers reported in the thesis can easily be read too literally if calibration is ignored.

The held-out protocol is therefore not a technical footnote. It is one of the main scientific stakes of the project. Training, internal cross-validation, calibration fitting, and tuning all occur inside the training process. Final claims are reserved for the frozen held-out test partition. That design prevents the thesis from turning into a document that implicitly learns from the final evaluation set while still describing itself as held-out. It also keeps the final probability-quality claims interpretable. If calibration had been fit using held-out outcomes, the most important methodological claim of the thesis would collapse immediately.

This is where the thesis differs from a shorter predictive paper. A shorter paper might report a summary AUC table and move on. A thesis of this scale has to show what the evaluation rule means, why it is stricter than default practice, and how that rule shapes the final interpretation. That added explanation is part of the scientific argument.

1.5 Why the Thesis Preserves a Strict Weighting Boundary

The weighting boundary belongs in the introduction because it is one of the earliest places where the thesis chooses scientific discipline over convenience. A simpler paper might use weighted descriptive results and predictive results interchangeably because both emerge from the same survey. This thesis does not. It treats weighting as part of the population-description task and held-out prediction as part of a different task. That choice avoids a large amount of conceptual drift later in the document.

The same boundary also explains why the thesis can remain formal without becoming confusing. Readers are asked to track two kinds of evidence, but each has a stable role. Weighted descriptive context explains why the problem matters. Unweighted held-out benchmarking determines whether the bullying block adds predictive value. The introduction states this early so later sections can become more technical without losing the logic of the overall design.

1.6 Research Question, Analytic Expectation, and Contribution

The central research question is whether QN24 and QN25 add incremental predictive value for QN26 after demographic adjustment under a fixed held-out protocol. The hypothesis, stated in the proposal and preserved throughout the study, is that the bullying block will improve held-out

probability quality and modestly improve discrimination while remaining strictly non-causal and research-stage in interpretation.

The main contribution is not a generic claim that bullying matters. Prior literature already establishes that. The contribution here is a calibrated, tightly scoped estimate of how much two bullying indicators improve held-out prediction beyond a narrow demographic baseline, together with a design that keeps weighting, prediction, subgroup review, interpretability, and governance in clearly separated roles. Conceptually, the thesis contributes a disciplined template for survey-based prediction studies in which descriptive surveillance context, probability estimation, subgroup auditing, and governance are treated as distinct but coordinated analytic tasks rather than as one blended claim.

That contribution is strengthened by the breadth of supporting analyses developed during the study. The thesis draws on missingness summaries, comparison tables, calibration outputs, stability checks, and structured documentation prepared during the research process. In the revised manuscript, the main text keeps only the evidence needed to sustain the central scientific argument, while the broader diagnostic archive is preserved in appendices so readers can audit the full chain of support without losing the pacing of the main results.

2. Literature Review

2.1 YRBS Surveillance Context and the Literature Boundary

The literature foundation for this thesis begins with the surveillance context itself. Brener et al. (2024) document the 2023 YRBSS methods and emphasize that the system is built for population surveillance. The Centers for Disease Control and Prevention (2024) provide the user guidance for the 2023 public-use data. These sources do more than establish credibility. They define the measurement and interpretation boundaries that any defensible secondary analysis must respect. The current thesis adopts those boundaries literally. Item definitions are preserved from the YRBS coding system, survey-design variables are retained for descriptive use, and the document resists any temptation to treat a public-use surveillance dataset as though it were a clinical registry or an intervention platform.

That boundary has a direct methodological implication. Much of the public health literature uses YRBS data to estimate weighted prevalence or adjusted associations. Those uses are appropriate for the data source, but they do not automatically determine how a thesis should structure a predictive comparison. The present thesis therefore treats the surveillance literature as a source of measurement discipline and substantive importance, not as a template for collapsing the project back into one more association paper.

2.2 YRBS-Based Mental Health and Bullying Research

The 2023 YRBS literature gives the thesis a strong substantive reason to focus on bullying and persistent sadness or hopelessness together. Young et al. (2024) reported that frequent social media use and bullying victimization co-occur with persistent sadness or hopelessness and suicide risk in the national 2023 survey. Kreski et al. (2022) showed that bullying and suicidal behavior vary across demographic subgroups in national adolescent data. English et al. (2024) found that bullying and suicidality trends vary across demographic groups in an urban surveillance context as well. Taken together, these studies make clear that bullying is not a niche topic within adolescent mental health surveillance. It is one of the recurring signals around which both descriptive disparity work and broader mental health concern are organized. Bezie et al. (2025) provide additional systematic-review evidence that bullying victimization is associated with suicidal ideation among students, which reinforces the broader mental-health relevance of the bullying construct without changing this thesis's predictive scope.

The important point for the thesis, however, is not simply that bullying and poor mental health co-occur. It is that the literature already provides enough substantive grounding to make a predictive test reasonable. The thesis does not need to spend pages proving that bullying matters in general. What it needs is a cleaner articulation of the next question, which is whether the bullying indicators still add out-of-sample signal after the baseline covariate set is fixed.

2.3 Bullying and Cyberbullying as Distinct Predictive Indicators

The literature also supports keeping school bullying and electronic bullying distinct. Ye et al. (2023) show that bullying victimization is associated with depressive symptoms across a wide evidence base, but that broader synthesis does not erase the differences between exposure contexts. Grimland et al. (2025) focus specifically on cyberbullying victimization and serious adolescent risk outcomes in a cross-national study. Hinduja and Patchin (2025) argue that cyberbullying should be understood as a real source of psychological harm rather than as a negligible online extension of offline conflict. Agarwal et al. (2026) show that school-based bullying victimization remains tied to serious mental health outcomes in the 2023 YRBS. Nagata et al. (2025) also connect cyberbullying with later mental-health and substance-use outcomes in a prospective early-adolescent cohort, further supporting the decision to keep electronic bullying visible as its own indicator.

For the present thesis, that distinction has two methodological consequences. First, it justifies keeping QN24 and QN25 as separate indicators inside one bullying block rather than reducing them to a single blended exposure variable. Second, it supports the decision to test the block's incremental value without pretending that the thesis can adjudicate the precise mechanism behind each exposure form. The model is allowed to use the indicators as distinct predictive signals. The thesis is not allowed to turn their predictive use into a mechanistic story it cannot substantiate.

2.4 Limits of Association-Dominant Work

The core literature gap is easiest to see by contrast with the dominant mode of prior work. A large share of the literature ends with weighted disparities, adjusted odds ratios, or discussion of risk-factor associations. Those outputs are useful. They describe burden, disparity, and observed association patterns. They do not answer whether a predictor block improves out-of-sample prediction after adjustment for a defined baseline. A predictor can be strongly associated with an outcome yet add little held-out information once baseline covariates are present. A predictor can also add held-out information even when causal interpretation remains unresolved.

This is the thesis boundary that matters most. The document is not trying to refute association-focused work. It is trying to show that a different evidentiary question can be asked of the same surveillance domain. The claim is narrower and, for that reason, more defensible.

2.5 Prediction-Evaluation Standards

The methodological literature gives direct support for that shift in evidentiary standard. Efthimiou et al. (2024) argue that prediction studies require disciplined development, validation, and reporting rather than ad hoc performance summaries. Cai et al. (2025) motivate the need for

uncertainty-aware interpretation of validation results, especially when performance summaries can move with data partitioning or resampling. These sources help define what a thesis-grade predictive study must report if it wants to be taken seriously. It cannot stop at one accuracy-style table. It must explain how the final comparison was structured, how tuning and calibration were separated from held-out evaluation, and how stability was judged.

The methodological demands become even clearer once the thesis is understood as probability-focused. If the outputs being interpreted are probabilities rather than only class labels, then the quality of those probabilities has to be evaluated directly. This is where the calibration literature enters the design, not as an advanced add-on but as a necessary consequence of the way the thesis poses its own question.

2.6 Calibration as Co-Primary Evidence

Huang et al. (2020) and Silva Filho et al. (2023) argue that calibration should be reported alongside discrimination when predicted probabilities are interpreted. That principle is central to the present thesis. The document is not content to ask whether one model ranks cases better than another. It asks whether the reported probabilities are better scaled and more defensible once the bullying block is added and once the model family is selected. Brier score, calibration slope, calibration intercept, and a probability-bin diagnostic therefore become co-primary or near-primary evidence rather than decorative diagnostics.

This is an important departure from a common pattern in which AUC carries the main argumentative burden and calibration is reported only superficially. In this thesis, calibration changes the model-selection story. The final candidate wins not because it posts a dramatic discrimination edge, but because it offers the best probability-quality balance once the document's lexicographic rule is applied. That would not be visible in a discrimination-only framework.

2.7 Subgroup Review and Bounded Interpretability

The literature also points toward subgroup review and bounded interpretability as design requirements rather than optional extras. Kreski et al. (2022) and English et al. (2024) show that bullying and adolescent mental health outcomes vary across demographic groups. In a predictive setting, that observation does not automatically imply unfairness or unusability. It does imply that average performance can hide meaningful heterogeneity. A thesis that never examines subgroup behavior would therefore leave an obvious gap in its evidentiary structure.

Interpretability poses a parallel challenge. Saeed and Omlin (2023) caution against treating explainability methods as substitutes for rigorous modeling and reporting. That warning is highly

relevant here. The thesis uses interpretation tools as secondary checks on model behavior rather than as independent proof of causal or mechanistic relationships.

2.8 Governance and Research-Only Framing in the Literature

The literature and policy environment also support a strong governance boundary. The Office of the Surgeon General (2023) frames digital exposures and youth mental health as part of a wider risk environment, not as a justification for oversimplified automated operational responses. That framing aligns well with the present thesis. The thesis treats its findings as research-stage predictive associations, not as deployment-ready outputs. This governance language follows directly from using public-use, cross-sectional, self-reported survey data about adolescent mental health.

That governance boundary also protects the thesis from the most common overclaim in student machine-learning work. A held-out metric improvement is not evidence that a model should be used in practice. The literature supports that restraint. The thesis makes it explicit.

2.9 Why YRBS Surveillance Is an Awkward but Valuable Prediction Setting

The literature also implies that YRBS is an awkward but valuable setting for prediction. It is awkward because it was not built as a prediction registry. The data are self-reported, the survey is cross-sectional, and the public-use release compresses the world into a finite set of coded items. At the same time, it is valuable because the survey has national relevance, standardized measurement, and public health salience. That combination makes it ideal for a careful research-stage prediction study as long as the thesis does not confuse the strengths of the survey with strengths it does not have.

This awkwardness is not a weakness to be hidden. It is part of what makes the thesis more serious. The thesis can use the YRBS productively only by acknowledging that the survey is stronger as a descriptive and surveillance instrument than as a deployment substrate. Once that is admitted, the value of a calibration-first, held-out, research-only design becomes much clearer.

2.10 Why the Literature Implies a Narrow Baseline and Explicit Governance Boundary

The literature review also supports the narrowness of the baseline and the explicitness of the governance language. Substantive studies show that bullying matters. Methodological studies show that calibration and reporting discipline matter. Policy and governance sources show that adolescent mental health prediction should be discussed carefully. Put together, those literatures point toward a thesis that is tighter rather than looser. They support a design in which the

baseline is defined in advance, the bullying block is tested explicitly, and the final language remains non-causal and research-only.

This point is important because the thesis is deliberately resisting two forms of sprawl. It resists empirical sprawl, which would add many more predictors and lose the clarity of the bullying-block comparison. It also resists rhetorical sprawl, which would let a modest but real predictive result mutate into a broad causal or operational claim. The literature itself provides the reason for that restraint.

2.11 Literature-to-Design Synthesis

Taken together, the literature supports a specific thesis design. The YRBS methods sources justify tight measurement discipline. The bullying and mental health literature justifies focusing on QN24, QN25, and QN26. The predictive-modeling literature justifies calibration-centered evaluation, subgroup review, and explicit interpretive limits. The result is a design that is narrow enough to be coherent and rich enough to be scientifically interesting.

The literature review therefore ends with a methodological bridge rather than with a generic summary. Prior work makes the present predictive question both interesting and legitimate. It does not answer that question. The remaining sections translate the gap into a fixed data source, a fixed feature system, a fixed evaluation rule, and a fixed set of robustness and governance boundaries. That translation is what allows the thesis to function as an integrated research paper rather than a stitched set of assignments.

3. Methods

3.1 Data Source, Survey Context, and Analytic Sample

This subsection establishes the local data posture, the survey design context, the analytic sample, and the descriptive boundaries that structure every later model comparison.

3.1.1 Public Data Source and Local Data Posture

This thesis uses local 2023 National Youth Risk Behavior Survey public-use source files to construct a reproducible modeling table. The raw source workbook remains local so the study does not redistribute source microdata unnecessarily. The analysis instead preserves the scripts, configuration details, tables, figures, and documentation needed to reconstruct the analysis-ready table. That division between local raw-data handling and documented analytical outputs is intentional. It protects the study's data posture while still allowing the analytic sequence to be checked and reproduced.

The decision to build the thesis around documented outputs rather than raw file redistribution matters for the shape of the document. The thesis can report exact counts, coding boundaries, and modeling decisions because those are captured in generated tables and supporting documents. It does not need to insert raw data extracts into the thesis. This keeps the main text focused on measurement and design logic rather than on low-level data handling detail.

3.1.2 Survey Design Context Relevant to the Predictive Task

The YRBS is a complex survey with design variables weight, stratum, and psu. Those fields matter because they define the descriptive population frame used throughout public health reporting. They also matter because they create an immediate design choice for the thesis. The project preserves the design variables for descriptive work and for documentation of the underlying survey context. It does not use them as predictive inputs. This boundary is not an implementation accident. It is part of the research design.

The survey design context has a second consequence as well. The public-use dataset is cross-sectional and self-reported. That means the thesis can study observed predictive relationships inside this survey context, but it cannot adjudicate temporal ordering, intervention effects, or clinical deployment value. The survey design therefore shapes not only the descriptive weighting boundary but also the interpretive ceiling of the manuscript. The manuscript is stronger when those ceilings are stated clearly rather than softened through optimistic rhetoric.

3.1.3 Analytic Sample Construction

The raw source file contains 20,103 rows. The final analysis-ready modeling table contains 19,863 rows and 14 columns. The difference between those counts comes from one deliberately narrow exclusion rule. Rows were retained if the primary outcome QN26 could be coded into the

binary target `y_qn26`. Rows without usable outcome coding were excluded before any final comparison work. This removed 240 rows and left one stable analytic sample for all later model comparison, calibration, and robustness work.

This sample construction strategy is important because it avoids a moving denominator across candidate models. The thesis is not comparing models built on different row subsets because one feature happens to have more missingness than another. Instead, it defines one shared modeling table and uses stable preprocessing within that table. That makes the held-out bullying-block comparison cleaner and easier to interpret. Any metric differences between the comparator and the expanded model can be read as design differences within one common analytic sample rather than as outputs of drifting sample eligibility.

Table 1. *Analytic Sample Construction and Modeling Boundary*

Sample-construction step	Rows	Interpretive role
Raw public-use source rows	20103	Starting point before thesis-specific outcome eligibility filtering
Rows removed for missing primary outcome	240	Outcome-based exclusion only
Final analysis-ready modeling rows	19863	Shared sample for all locked comparisons
Final analysis-ready columns	14	Outcome, bullying block, baseline covariates, and design fields
Positive outcome count	8108	Number of <code>`y_qn26 = 1`</code> rows in the final modeling table
Negative outcome count	11755	Number of <code>`y_qn26 = 0`</code> rows in the final modeling table
Unweighted outcome rate	0.408196	Outcome prevalence inside the modeling table

3.1.4 Measurement Properties Relevant to the Predictive Task

Several measurement features of the public-use dataset matter directly for the predictive task. First, the core thesis variables are self-reported items rather than administrative or clinical measurements. That means the thesis is modeling reported experience and reported mental health burden as captured by the YRBS instrument. Second, the project retains the YRBS item boundaries rather than redefining the constructs through custom composite scales. This keeps the final predictive question tightly aligned with the public-use measurement system.

Third, the predictor set is intentionally sparse. The final modeling table contains the outcome, two bullying indicators, four demographic covariates, and the survey-design variables preserved

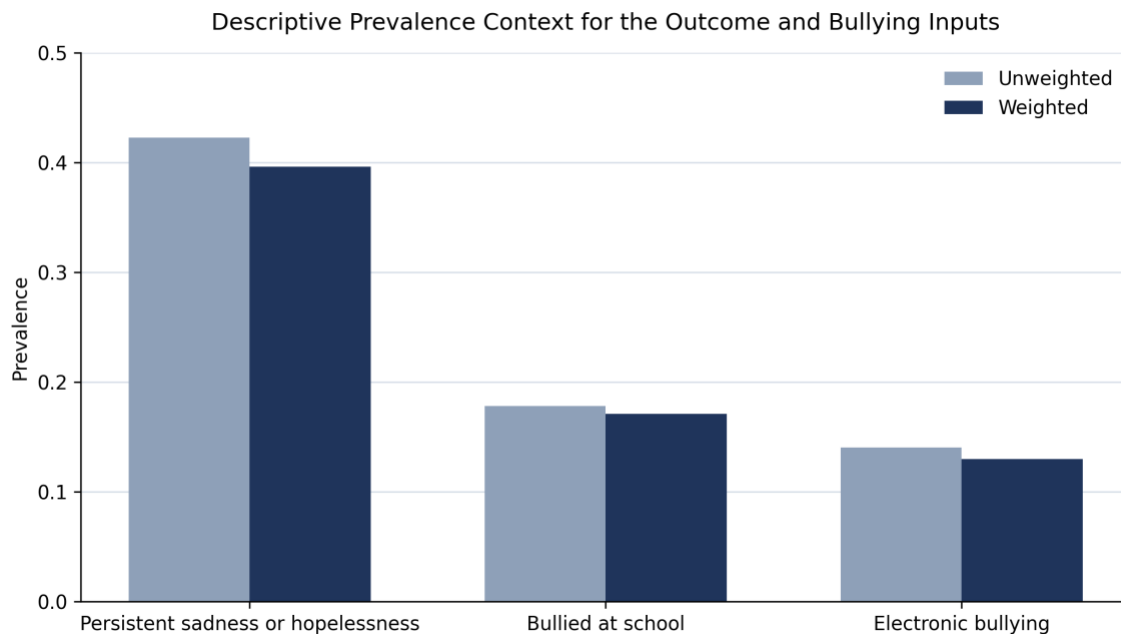
for descriptive work. The compactness of this table is a methodological strength. It makes the bullying-block comparison interpretable, reduces scope drift, and prevents the thesis from quietly turning into a broad feature-hunting exercise that the approved proposal never promised. A thesis can be long and exact without becoming indiscriminate.

3.1.5 Weighted Descriptive Anchors and the Predictive Boundary

The thesis uses weighting for descriptive context only. Weighted prevalence estimates show why the thesis question matters at the population level. They do not determine the predictive claim. In the local outputs, weighted prevalence is 39.6 percent for QN26, 17.1 percent for QN24, and 13.0 percent for QN25. These values show that the target outcome is common and that the bullying indicators are prevalent enough to serve as meaningful predictive inputs rather than as rare edge-case signals.

At the same time, the predictive design uses unweighted held-out benchmarking. That separation is deliberate. Weighted prevalence is a population-descriptive quantity. Held-out discrimination, Brier score, and calibration are model-evaluation quantities. Keeping those roles distinct prevents the thesis from mixing descriptive representativeness with predictive fitting in ways that would blur the central design question.

Figure 1. *Weighted and unweighted prevalence for the primary outcome and the two focal bullying indicators. The figure provides descriptive context only and does not determine held-out model selection.*



3.1.6 Descriptive Heterogeneity Across Age and Race or Ethnicity Proxies

The thesis still needs a limited statement about subgroup descriptive heterogeneity before the predictive sections become fully technical. The weighted descriptive outputs show that the

prevalence of QN26 varies across the age proxy and race or ethnicity proxy fields preserved in the analysis-ready table. That heterogeneity does not determine the model and does not substitute for held-out subgroup review. It does, however, explain why subgroup review later becomes a methodological requirement rather than an optional side analysis.

Across the age proxy categories with meaningful sample size, weighted QN26 prevalence ranges from roughly 38.0 percent to 42.9 percent. Across the more stable race or ethnicity proxy categories, weighted QN26 prevalence ranges from roughly 25.8 percent to 49.7 percent. These descriptive differences are important because they show the population context in which the predictive task sits. They do not, by themselves, determine the final predictive comparison.

Those subgroup descriptive anchors are preserved later in the results and appendix materials rather than as early body figures. That placement keeps the motivation visible while preventing the opening methods pages from carrying more subgroup detail than the main argument requires.

3.1.7 Missingness Profile and Outcome Eligibility

Missingness in the final modeled variables is low. In the analysis-ready table, `y_qn26` has no missing values because outcome missingness defines the sample filter. Missingness is 0.79 percent for `x_qn24`, 0.74 percent for `x_qn25`, 0.40 percent for `q1`, 0.73 percent for `q2`, 0.82 percent for `q3`, and 1.77 percent for `raceeth`. These values support the use of a simple and stable preprocessing strategy rather than a dedicated missing-data model. They also reinforce why one stable modeling table is feasible.

Low missingness does not mean measurement is perfect. It means the thesis does not need to convert itself into a missing-data methods study to produce a defensible predictive comparison. The observed missingness pattern is sparse enough that the body can report it, explain the preprocessing choice, and move on to the feature-system question that actually defines the thesis.

Figure 2. Missingness profile for the analysis-ready variable set. The figure supports the decision to use a stable low-complexity preprocessing strategy rather than a dedicated missing-data model.

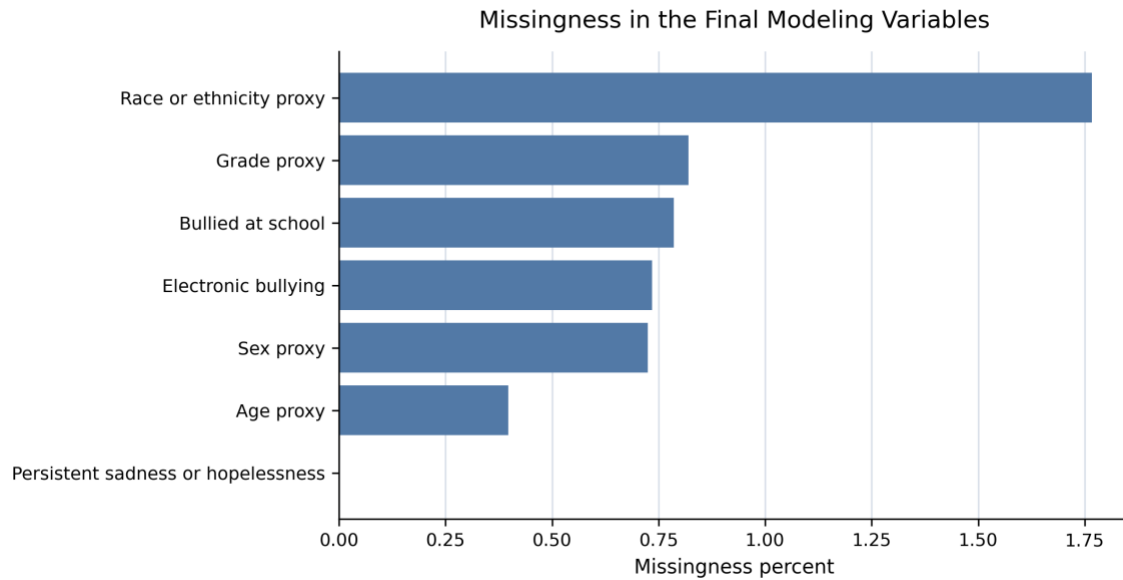


Table 2. Focused Missingness for Final Modeling Variables

Variable	Missing count	Missing rate	Design implication
`y_qn26`	0	0.0000	Sample filter yields complete outcome coding in the final table
`x_qn24`	156	0.0079	Low enough for stable simple imputation handling
`x_qn25`	146	0.0073	Low enough for stable simple imputation handling
`q1`	79	0.0040	Low enough for stable simple imputation handling
`q2`	144	0.0073	Low enough for stable simple imputation handling
`q3`	163	0.0082	Low enough for stable simple imputation handling
`raceeth`	351	0.0177	Highest among the core modeled variables but still modest

3.1.8 Why Survey-Design Variables Are Preserved but Excluded from the Predictive Feature Set

The survey-design variables weight, stratum, and psu are preserved exactly because they are part of the YRBS descriptive architecture. They are excluded from the predictive feature set because they are not the substantive student-level signals the thesis is designed to compare. Including them as predictive inputs would blur the boundary between descriptive survey mechanics and predictor-block comparison. It would also weaken the interpretability of the baseline versus bullying-block contrast by allowing sampling-design outputs to enter the model as though they were meaningful exposures or covariates.

This choice also protects the weighting boundary. The thesis uses the survey-design variables to describe the population context and to explain why the public-use dataset must be handled carefully. It does not use them to inflate predictive performance. That is a stronger design choice because it avoids a more ambiguous modeling strategy that could blur the line between survey mechanics and substantive predictors.

3.1.9 Cross-Sectional Implications for Prediction

Using a public-use cross-sectional dataset has substantive and methodological consequences. It means the thesis can speak about predictive association within an observed joint distribution, but it cannot establish temporal precedence or causal mechanism. It also means that calibration and held-out performance are being judged inside one survey wave rather than across time or across institutions. That is why the robustness section later emphasizes bounded confidence rather than generalization in the strong sense.

The cross-sectional nature of the data also sharpens the need for non-causal language. When the bullying block improves held-out prediction, the thesis is allowed to say that the indicators carry incremental predictive information in this design. It is not allowed to say that bullying exposure causes the observed risk surface or that a model based on these data should be operationalized. The measurement framework and the governance language therefore work together. They impose the same restraint from two different directions.

3.1.10 Section Summary

This section establishes the foundation for the rest of the thesis. It shows what the data source is, how the analytic sample was constructed, why weighting is treated descriptively rather than predictively, how low missingness was handled, and why the public-use survey context imposes real interpretive ceilings. These points are not background filler. They are part of the empirical architecture of the thesis. The next section turns that foundation into a formal feature system and a formal prediction task.

3.2 Data and Variable Design

This subsection defines the thesis-facing variable system, the baseline and bullying blocks, the preprocessing logic, and the leakage-safe split protocol used throughout the thesis.

3.2.1 Outcome Construction

The primary outcome is the binary target `y_qn26`, derived from the YRBS item QN26 persistent sadness or hopelessness. The coding rule is intentionally simple. Responses coded as the affirmative category are mapped to 1. Responses coded as the negative category are mapped to 0. Special missing codes are left missing and trigger exclusion before the modeling table is finalized. This direct recode preserves fidelity to the source instrument while making the target explicit enough for the predictive task.

Keeping the outcome construction simple is not a missed opportunity. It is a design choice that improves transparency. The thesis does not need a latent construct or a derived mental-health index to answer its central question. It needs a clearly documented outcome that is substantively meaningful, measurable in the 2023 YRBS, and compatible with the held-out protocol.

3.2.2 Bullying Predictor Block Construction

The bullying predictor block contains two binary indicators. `x_qn24` represents being bullied on school property. `x_qn25` represents being electronically bullied. Each is coded from the public-use YRBS item using the same yes or no mapping used for the outcome. Missing values are retained for preprocessing rather than serving as exclusion gates. This keeps the project aligned with one stable analytic sample and prevents the bullying block from changing the row universe it is supposed to be compared within.

Treating the two indicators as a predictor block rather than as one merged exposure preserves the substantive distinction supported by the literature. School-property bullying and electronic bullying may operate through different settings or reporting dynamics. The thesis does not need to resolve those mechanisms in order to use the variables predictively. It needs only to preserve their distinction and test their incremental value together under a fixed comparison rule.

3.2.3 Baseline Covariate Rationale

The locked baseline covariate set contains `q1`, `q2`, `q3`, and `raceeth`, corresponding to age, sex, grade, and race or ethnicity. This set is intentionally narrow. It is wide enough to constitute a meaningful demographic adjustment layer and narrow enough to keep the thesis centered on the incremental value of the bullying block. Broader covariate expansion was not pursued because it would have changed the scientific question from whether bullying indicators add value beyond a defined demographic baseline to whether they add value beyond an open-ended feature universe.

That choice deserves emphasis because it explains both the strength and the limit of the final claim. The strength is interpretability. The baseline versus bullying-block contrast is clean and transparent. The limit is scope. The thesis cannot claim that bullying remains incrementally predictive after every plausible covariate expansion. It can claim that the increment remains visible under the deliberately narrow demographic baseline that the study pre-specified and preserved.

Table 3. *Feature-System Design and Analytic Roles*

Component	Repository variable(s)	Thesis-facing label	Role in the predictive design
Primary outcome	`y_qn26`	Persistent sadness or hopelessness	Binary target for all supervised comparisons
Bullying indicator	`x_qn24`	Bullied on school property	Part of the incremental bullying block
Bullying indicator	`x_qn25`	Electronically bullied	Part of the incremental bullying block
Baseline covariate	`q1`	Age	Part of the locked demographic baseline
Baseline covariate	`q2`	Sex	Part of the locked demographic baseline
Baseline covariate	`q3`	Grade	Part of the locked demographic baseline
Baseline covariate	`raceeth`	Race or ethnicity	Part of the locked demographic baseline
Design variable	`weight`	Survey weight	Weighted descriptive context only
Design variable	`stratum`	Survey stratum	Weighted descriptive context only
Design variable	`psu`	Primary sampling unit	Weighted descriptive context only

3.2.4 Coding and Preprocessing Logic

The final preprocessing logic is intentionally low-complexity. Binary and categorical inputs are treated as categorical features. Sparse missingness is handled through stable simple imputation within the preprocessing pipeline. For the retained variables, this means that the model families operate on one consistent encoded feature space without changing the analytic sample. The project does not adopt a more elaborate preprocessing scheme because the missingness burden is low and because the central thesis question concerns the value of a fixed predictor block rather than the value of increasingly complex preprocessing.

This preprocessing discipline serves a second purpose. It keeps the model comparison focused on the difference between feature sets and model families rather than on undocumented data preparation tricks. A thesis can be technically serious without rewarding complexity for its own sake. In this project, serious design means keeping the coding pathway short, explicit, and repeatable.

3.2.5 Leakage Prevention and Frozen Split Protocol

The predictive analytic sequence uses a frozen split protocol anchored at seed 2026. The full analysis-ready table is partitioned into a training set and a held-out test set, with five cross-validation folds inside the training partition. Preprocessing, tuning, and calibration fitting occur only within the training process. The held-out test partition is reserved for final scoring and reporting. This is the main leakage-prevention rule of the thesis.

Leakage prevention matters especially because the final candidate uses post hoc calibration. Platt scaling and isotonic regression are both fit using training-derived predictions only. If the held-out outcomes had been used to fit the calibrator, the thesis could no longer claim that calibration quality was evaluated honestly on held-out data. The split protocol therefore protects both the discrimination results and the probability-quality results.

Table 4. Frozen Split Protocol and Analysis Roles

Object	Definition in this thesis	What it is allowed to do
<code>`D_train`</code>	Training partition after the fixed 80 to 20 split	Preprocessing, tuning, cross-validation, and calibration fitting
<code>`D_cv^(k)`</code>	Fold-specific validation slice inside <code>`D_train`</code>	Internal model comparison and stability summaries
<code>`D_test`</code>	Frozen held-out test partition	Final reporting only
Split seed	<code>`2026`</code>	Locks the primary train test partition for the thesis
Cross-validation folds	<code>`5`</code>	Provides within-training comparison structure
Calibration fitting boundary	Training-derived predictions only	Prevents information leakage from held-out outcomes

3.2.6 Formal Prediction Task Statement

Let each student observation be indexed by $i = 1, \dots, n$. Let y_i denote the binary QN26 outcome for student i . Let b_i denote the locked baseline covariate vector, u_i the two-variable bullying block, and x_i the combined feature vector. The central predictive question of the thesis can then be written as a comparison between a model using b_i only and a model using $x_i = (b_i, u_i)$.

$$\mathbf{b}_i = (q1_i, q2_i, q3_i, race_{eth_i}), \quad \mathbf{u}_i = (x_{24,i}, x_{25,i}), \quad \mathbf{x}_i = (\mathbf{b}_i, \mathbf{u}_i).$$

The baseline thesis question is whether the mapping from $\mathbf{b}_i$ to predicted outcome probability is materially improved when $\mathbf{u}_i$ is added, with all claims evaluated on the frozen held-out set rather than on the training process. This can be expressed compactly as a held-out comparison of feature-block-specific predictors.

$$f_B : \mathbf{b}_i \mapsto \hat{p}_{B,i}, \quad f_X : \mathbf{x}_i \mapsto \hat{p}_{X,i}.$$

Table 5. Core Notation and Mathematical Components

Symbol or object	Definition	Thesis role
y_i	Binary QN26 outcome for student i	Main supervised target
$\mathbf{b}_i$	Baseline covariate vector	Non-bullying comparator feature block
$\mathbf{u}_i$	Bullying predictor vector	Incremental block under test
$\mathbf{x}_i$	Full feature vector $(\mathbf{b}_i, \mathbf{u}_i)$	Input to the expanded models
$\hat{p}_i$	Predicted probability that $y_i = 1$	Probability output used for evaluation
$\mathcal{D}_{\text{train}}$	Training partition	Training-side learning and calibration only
$\mathcal{D}_{\text{cv}}^{(k)}$	Fold-specific internal validation slice	Cross-validation comparison object
$\mathcal{D}_{\text{test}}$	Frozen held-out test partition	Final scoring object
f_B	Baseline comparator model	Tests the baseline block alone
f_X	Expanded model with bullying block	Tests incremental value beyond the baseline

3.2.7 Logistic Reference Model

The first candidate family is regularized logistic regression. Its role in the thesis is twofold. It provides a transparent reference model and it gives the thesis a clear probability model against which the nonlinear family can be judged. For the full feature vector $\mathbf{x}_i$, the logistic probability model is

$$\hat{p}_i = P(Y_i = 1 \mid \mathbf{x}_i) = \sigma(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}) = \frac{1}{1 + \exp[-(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})]}.$$

The logistic family matters even when it does not win the final model-selection rule. It establishes whether a relatively simple linear probability surface already captures most of the available signal. Without that reference, the nonlinear model family would be harder to interpret. The thesis would know only that one model performed well, not whether nonlinear flexibility actually bought enough to justify its use.

3.2.8 Regularized Objective

The fitted logistic reference uses regularization to stabilize estimation and preserve generalization under the compact feature system. A thesis-appropriate form of the regularized empirical objective is

$$\mathcal{L}(\beta_0, \boldsymbol{\beta}) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] + \lambda \|\boldsymbol{\beta}\|_2^2.$$

The point of writing the objective explicitly is not performative mathematics. It is to make clear what the logistic family means in this thesis and why it is a sensible reference model. The logistic family estimates one smooth probability surface from the encoded predictors and penalizes excessive coefficient magnitude. That is enough formal detail for the thesis without turning the methods chapter into a general treatment of penalized optimization.

3.2.9 Histogram Gradient Boosting Formulation

The second candidate family is histogram gradient boosting. This family is included because it can capture nonlinearities and interactions while still being evaluated by the same held-out probability-quality criteria as logistic regression. A concise thesis-facing formulation is to represent the score function as a stage-wise additive sum of shallow trees:

$$\eta_M(\mathbf{x}) = c + \sum_{m=1}^M v h_m(\mathbf{x}), \quad \hat{p}_M(\mathbf{x}) = \sigma(\eta_M(\mathbf{x})).$$

Here h_m denotes the stage-specific tree increment and v the learning-rate factor. This is the level of mathematical seriousness the thesis needs. It states what the boosting family is doing without pretending that the thesis needs a full derivation of tree-splitting mechanics. The important substantive point is that the nonlinear family can produce a better probability surface than the linear reference while still being judged by the same held-out criteria.

3.2.10 Candidate Family Rationale and Non-Pursued Expansions

The final candidate set is deliberately compact. Logistic regression offers a transparent reference and a useful calibration baseline. Histogram gradient boosting offers controlled nonlinear

flexibility. More expansive family searches were not pursued because they would have shifted the thesis away from an interpretable block-comparison design and toward a more open-ended model tournament. That shift would have made the final claim broader but less exact.

The same logic applies to feature expansion. The project did not attempt to add every available survey variable, interaction, or latent feature representation. The goal was not to maximize model complexity. The goal was to determine whether the bullying block adds measurable predictive value beyond a pre-specified baseline. The candidate-family rationale and the narrow feature-set rationale therefore reinforce one another. Both are acts of design discipline.

Table 5A. *Deferred Feature and Model Expansions in the Locked Thesis Design*

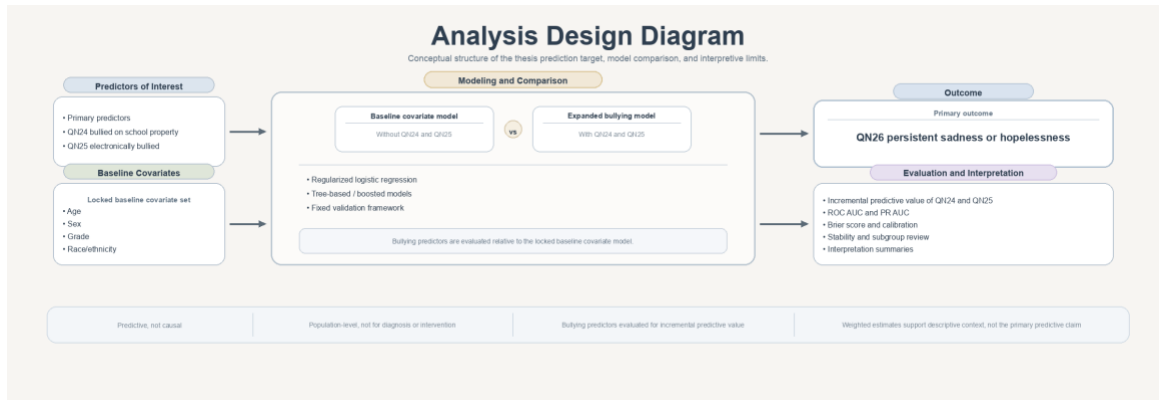
Deferred expansion	Why it was not pursued in this thesis	What would be required to justify it later
Broader survey covariate expansion	Would change the meaning of the baseline comparator and weaken the clarity of the bullying-block question	A new protocol with a different comparator definition
Interaction-rich manual feature engineering	Would move the project toward post hoc feature hunting rather than a locked block comparison	A pre-specified interaction design and separate validation plan
Additional model-family tournament	Would emphasize leaderboard breadth over probability-quality interpretation	A broader study whose objective is model-family benchmarking rather than thesis-native design clarity
External feature linkage	Not available in the local project materials and would exceed the public-use thesis posture	A separate data-governance and linkage analytic sequence

3.2.11 Why Broader Covariate Expansion Was Deliberately Deferred

The decision not to broaden the covariate system further deserves its own explicit defense. The study materials make a larger predictor universe imaginable, but the thesis would become less exact if it quietly expanded the baseline until the bullying-block comparison lost its clean meaning. A common failure mode in student prediction work is to let the feature set grow until the central contrast no longer has an identifiable scientific role. This thesis resists that drift.

This matters because the final result depends on the interpretability of the comparator. When the non-bullying comparator is the locked demographic block, the reader knows exactly what is being held fixed. If the comparator instead became a wide, opportunistic feature set assembled late in the project, then the incremental value of the bullying block would become harder to parse and easier to overstate. The thesis gains coherence by refusing that drift.

Figure 3. Analysis design diagram summarizing the locked outcome, the narrow baseline block, the bullying block, the held-out protocol, and the non-causal interpretation boundary.



3.2.12 Why the Baseline Is Intentionally Narrow

It is worth stating the narrow-baseline rationale one more time because it is easy to misread. The baseline is not narrow because additional covariates are unimportant. It is narrow because the thesis is asking a clean block-comparison question. A broader baseline would convert the study into a different project, one in which the bullying block is judged against a much larger and more heterogeneous set of contextual features. That could be a good future study. It is not the current one.

The narrow baseline also protects the main empirical interpretation. If the final thesis reported dozens of predictors but still claimed that the bullying block was the central contribution, the reader would have good reason to doubt that the thesis question had remained stable. By keeping the baseline compact, the thesis makes the bullying-block increment legible and keeps the final results aligned with the approved protocol.

3.2.13 Section Summary

This section formalizes the feature system that the prior methods discussion only motivated. It defines the outcome, the baseline block, the bullying block, the split protocol, the candidate families, and the exact prediction rule that governs the central empirical comparison.

3.3 Evaluation, Calibration, and Model Selection

This subsection formalizes how candidate models are evaluated, calibrated, and selected under a held-out, probability-first decision rule.

3.3.1 Held-Out Evaluation Philosophy

The evaluation framework of this thesis is held-out and probability-first. The thesis does not treat the modeling problem as a competition in which a model wins because it posts the largest single discrimination number. Instead, it asks whether the model produces better predicted probabilities under a fixed evaluation rule. This philosophy is not cosmetic. It governs how candidates are ranked, how calibration is interpreted, and how the final bullying-block increment is understood.

That emphasis on held-out evaluation matters because the document is meant to read like a research thesis rather than like a development notebook. Internal cross-validation results are useful because they show how candidate models behave inside the training process. They are not the main claim. The main claim rests on the frozen held-out test partition. This simple distinction becomes a major source of document authority once it is stated clearly and followed consistently.

3.3.2 Why Discrimination Alone Is Insufficient

Discrimination metrics remain important. ROC AUC and PR AUC tell the reader how well a model ranks positive cases above negative cases across threshold choices. But discrimination is insufficient when the thesis is explicitly reporting predicted probabilities and using those probabilities to compare models. A model can rank cases reasonably well while still producing probabilities that are too extreme, too compressed, or systematically shifted. In a setting like adolescent public health prediction, that probability scaling problem is not an edge case. It is one of the main reasons to evaluate calibration directly.

That is why the thesis does not let discrimination dominate the narrative. If it did, the final HGB candidates would appear almost interchangeable and the thesis would miss the main methodological point of its own design. Brier score and calibration are what make the final model-selection argument more mature than an AUC leaderboard.

Table 6. *Evaluation Quantities and Decision Roles*

Quantity	Formal object	What it measures in this thesis	Decision role
ROC AUC	Rank-based discrimination summary	Ability to rank positives above negatives	Third lexicographic tie-breaker
PR AUC	Precision-recall summary	Discrimination with attention to class imbalance	Secondary descriptive support

Quantity	Formal object	What it measures in this thesis	Decision role
Brier score	Mean squared probability error	Overall probability quality	First lexicographic criterion
Calibration slope	Logistic recalibration slope	Whether probabilities are too compressed or too extreme	Second lexicographic criterion
Calibration intercept	Logistic recalibration intercept	Average overprediction or underprediction	Companion calibration descriptor
Probability-bin gap	Observed minus predicted within bins	Localized calibration interpretation	Supplementary diagnostic
Bootstrap interval	Percentile interval over held-out resamples	Uncertainty around reported held-out summaries	Robustness support

3.3.3 Calibration-Centered Decision Logic in Concrete Terms

The lexicographic rule is easier to understand when the thesis states the kinds of candidate conflicts it is designed to resolve. The final decision rule is not merely a ranking of metrics. It is a statement about what kinds of model behavior the thesis is willing to privilege. A model that improves ROC AUC slightly but worsens Brier score is not attractive under this design. A model that preserves Brier score while materially compressing calibration slope is also not attractive.

This concrete framing matters because the thesis compares candidates that are already fairly strong. It is easy to overread tiny differences when the underlying candidates are competent. The design logic in this section is meant to prevent that. The thesis does not reward whichever candidate looks best on the most visually dramatic figure. It rewards the candidate whose predicted probabilities are most defensible under the project's research-only aims.

Table 6A. *Lexicographic Decision Hierarchy and Candidate Failure Modes*

Candidate behavior	Why it is insufficient	How the thesis rule treats it
Slightly better ROC AUC with worse Brier score	Improves ranking while worsening probability quality	Rejected in favor of the lower-Brier candidate
Similar discrimination with slope farther from one	Leaves probability scaling less defensible	Rejected if the Brier tie is close and slope is materially worse
Better intercept but weaker discrimination and worse Brier score	Improves one calibration component while degrading the broader decision target	Rejected because the rule is not intercept-only
Similar discrimination with modestly better Brier and better slope	Improves the exact quantities the thesis prioritizes	Preferred under the thesis rule

3.3.4 Formal Metric Definitions

Let s_i denote the scalar score or probability assigned to observation i . The thesis uses the following core metric definitions. ROC AUC is treated as the probability that a randomly selected positive case receives a higher score than a randomly selected negative case:

$$AUC_{ROC} = P\{s(X^+) > s(X^-)\}.$$

PR AUC is the area under the precision-recall curve induced by the score ranking:

$$AUC_{PR} = \int_0^1 \text{Precision}(r) dr.$$

For a held-out set T with m observations, the Brier score is

$$\text{Brier} = \frac{1}{m} \sum_{i \in T} (\hat{p}_i - y_i)^2.$$

These definitions are enough for the thesis because they show exactly what is being optimized and interpreted. The study does not need longer derivations. It needs readable definitions that keep the evaluation section mathematically explicit without becoming ornamental.

3.3.5 Calibration Intercept and Slope

Calibration is assessed through a logistic recalibration model fit on held-out predictions for evaluation and on training-derived predictions for calibration fitting. The interpretive form of the recalibration model is

$$\text{logit}\{P(Y_i = 1)\} = \alpha + \gamma \text{logit}(\hat{p}_i).$$

Here α is the calibration intercept and γ is the calibration slope. An intercept near zero indicates that the average probability level is well centered. A slope near one indicates that the probabilities are neither too extreme nor too compressed. These quantities matter because they let the thesis distinguish between two models with almost identical discrimination but different probability behavior. That is exactly the situation encountered in the final HGB calibration comparison.

3.3.6 Probability-Bin Diagnostic Framework

The proposal required a probability-bin diagnostic, and the retained held-out prediction outputs are sufficient to support it. The thesis uses decile-style probability bins as an interpretive device in the main text. For bin g , let $\bar{p}_g$ be the mean predicted probability and $\bar{y}_g$ the observed event rate. The local calibration gap is then

$$\Delta_g = \bar{y}_g - \bar{p}_g.$$

Positive values indicate underprediction in that region of the score range. Negative values indicate overprediction. The thesis is careful not to elevate the bin table into a rival selection rule. Its job is explanatory. It shows what calibration error looks like in concrete score regions once the global summaries have already identified the main model-selection story.

3.3.7 Platt Scaling and Isotonic Regression

Post hoc calibration sensitivity is assessed using no recalibration, Platt scaling, and isotonic regression. If z_i denotes the raw model score, then Platt scaling fits a logistic map

$$\tilde{p}_i = \sigma(az_i + b).$$

Isotonic calibration instead fits a monotone nondecreasing function g to the raw model probabilities:

$$\tilde{p}_i = g(\hat{p}_i), \quad g \text{ monotone nondecreasing.}$$

In this thesis, both procedures are fit only on training-derived predictions. That boundary is essential. It keeps the held-out test set from leaking into the probability-scaling step and therefore preserves the integrity of the final calibration comparison.

3.3.8 Uncertainty Framing and Bootstrap Logic

Uncertainty is also reported through stratified held-out bootstrap intervals for the archived uncalibrated full HGB reference run that sits adjacent to the final calibration comparison. Let M denote a target metric, and let $M(1), \dots, M(B)$ denote the replicate values of that metric across B bootstrap samples drawn from the held-out set with outcome stratification. The percentile interval is

$$CI_{0.95}(M) = [q_{0.025}(M^{*(b)}), q_{0.975}(M^{*(b)})].$$

The bootstrap layer does not transform the thesis into an external-validation study. It gives the reader a bounded uncertainty frame around the main held-out summaries. That is enough for the thesis because the goal is not to overstate certainty. It is to avoid presenting point estimates as if they were self-sufficient.

3.3.9 Lexicographic Model-Selection Rule

The final model is selected lexicographically. Among the candidate set C , the winning model is chosen by minimizing held-out Brier score first, then the absolute distance of the calibration slope from one, and then by maximizing ROC AUC:

$$f^* = \operatorname{argmin}_{f \in C} (\text{Brier}(f), |y_f - 1|, -\text{AUC}_{\text{ROC}}(f)).$$

This rule is elegant because it states the scientific priorities of the thesis in one place. Probability quality comes first. Calibration stability comes second. Discrimination remains important but does not overrule clear probability-quality advantages. This is what makes the final HGB with Platt scaling choice defensible as a consequence of pre-stated priorities rather than of retrospective preference.

3.3.10 Why Brier Score Outranks PR AUC in This Exact Project

The project reports PR AUC because the outcome is not balanced and because precision-recall behavior remains informative. It does not place PR AUC above Brier score in the final lexicographic hierarchy. That ordering is intentional. PR AUC is still a discrimination summary. It says little about whether the probabilities themselves are well scaled. Brier score, by contrast, speaks directly to the quality of the predicted probabilities the thesis is choosing to emphasize.

This ordering is therefore not a general rule for every prediction problem. It is the right rule for this thesis because the manuscript has already committed itself to a probability-oriented interpretation. Once that commitment is made, Brier score has to outrank a second discrimination summary or the method section would be internally inconsistent.

3.3.11 Why Calibration Intercept Is Reported but Not a Primary Selector

Calibration intercept is reported prominently because it conveys average overprediction or underprediction. It is not used as a primary selector because intercept alone can be improved or worsened without solving the broader probability-quality problem that the thesis cares about. A model can have an intercept closer to zero and still be too compressed, too extreme, or worse on Brier score. The thesis therefore reports intercept as an important companion quantity rather than as a decisive target on its own.

That reporting choice helps the reader understand why the final selection rule is both mathematically explicit and still selective. The thesis is not ignoring calibration detail. It is distinguishing between calibration quantities that have different interpretive jobs.

3.3.12 Validation, Robustness, and Interpretability Are Distinct Analytic Layers

One conceptual risk in long technical documents is to let every supportive analysis collapse into a single vague idea of validation. The thesis avoids that. Validation refers to the core held-out

scoring framework. Robustness refers to multiseed, bootstrap, and hyperparameter-sensitivity checks. Subgroup review refers to group-specific performance summaries under explicit adequacy rules. Interpretability refers to permutation-based stability of the final feature-engagement story. Keeping those layers distinct prevents the thesis from promising more than the evidence can support.

Keeping these categories distinct improves both rigor and readability. It prevents the manuscript from sounding more certain than it is and gives each later section a clearer job. The next section uses that distinction to turn robustness and governance into a full methodological component of the thesis rather than a short coda attached to results.

3.3.13 Section Summary

This section provides the rulebook for the rest of the thesis. It defines the metrics, explains why calibration matters, formalizes the bin diagnostic, and makes the model-selection rule explicit. That rule is what later lets the thesis say exactly why the final candidate wins. The next section then extends that architecture by treating robustness, subgroup review, interpretability, governance, and reproducibility as part of the method rather than as afterthoughts.

3.4 Robustness, Stability, and Governance Constraints

This subsection explains the supplementary checks used to test multiseed stability, subgroup consistency, and research-use boundaries without weakening the locked primary comparison.

3.4.1 Robustness, Stability, and Scope Controls

Robustness and governance warrant their own section because the central thesis claim depends on more than a single held-out comparison. A model can win one held-out table and still be too brittle, too subgroup-dependent, or too interpretively unstable to support a strong research-stage conclusion. By separating robustness into its own section, the thesis makes clear that stability and restraint are part of the scientific design rather than post hoc reassurance.

This separation also improves the body design. Robustness evidence should not be buried in a detached diagnostic archive. Selected robustness logic and selected summary results belong in the main text so the reader can evaluate the strength and limits of the final result directly.

Table 7. Robustness Layers, Supporting Evidence, and Interpretive Limits

Robustness layer	Primary evidence object	What it contributes	What it cannot prove
Multiseed stability	Cross-validation summaries across seeds 2026 through 2029	Tests whether the selected family is fragile to nearby seed perturbation	External validity across new datasets
Held-out bootstrap intervals	Stratified bootstrap intervals for held-out metrics	Quantifies uncertainty around final held-out summaries	Transportability beyond the observed sample
Hyperparameter sensitivity	Local perturbation table around the tuned HGB setting	Checks whether the tuned setting sits on a visibly unstable surface	Global optimality across all settings
Subgroup review	Sex and race or ethnicity review table	Checks for obvious group-specific failures under defined thresholds	Normative fairness certification
Interpretation stability	Permutation-importance stability summary	Checks whether the bullying block remains consistently engaged	Causal explanation
Governance review	Deployment and use constraints plus risk register	Bounds the meaning and use of the findings	Safe operational deployment

3.4.2 Multiseed Stability Logic

The multiseed analysis examines whether the archived uncalibrated full HGB reference configuration behaves materially differently under nearby seed perturbations while preserving

the same held-out split seed. Let M_s denote a cross-validation metric summary at seed s . The thesis then asks whether the general scale of those metric summaries remains stable across nearby seed values.

$$\bar{M} = \frac{1}{S} \sum_{s=1}^S M_s, \quad \text{range}(M_s) = [\min_s M_s, \max_s M_s].$$

This logic matters because it frames the multiseed analysis correctly. The thesis is not searching across seeds for a better model. It is checking whether the archived reference configuration looks obviously unstable under small perturbations. That is the correct research-stage use of seed variation in this project.

3.4.3 Bootstrap Interval Logic

The bootstrap layer addresses a different concern. Even with a fixed held-out partition and a fixed archived reference configuration, the reported metrics are still sample-dependent summaries. Stratified bootstrap intervals help show how much those summaries might move under resampling of the held-out set. They are especially useful for the Brier score and calibration slope because those quantities are central to the thesis but can be difficult to interpret if reported only as single point estimates.

The bootstrap logic therefore deepens the probability-first orientation of the manuscript. It tells the reader not only what the archived held-out summaries are, but also how tightly they are bounded inside the observed held-out data. That is an important difference between a manuscript that foregrounds uncertainty and a shorter paper that might simply report the final figures without that context.

3.4.4 Hyperparameter Sensitivity Logic

Hyperparameter sensitivity addresses yet another concern, namely whether the archived tuned full HGB reference configuration sits on an unstable local surface. The retained tuning outputs include a small grid of local perturbations around that tuned point. The thesis uses those perturbations not to reopen tuning but to show that the full HGB reference family does not owe its behavior to an implausibly sharp configuration choice. When Brier score and related summaries remain within a narrow range under modest perturbations, the archived reference family looks substantially more credible.

This sensitivity layer is especially important because histogram gradient boosting can otherwise look like a family whose behavior depends on opaque tuning. The local perturbation table lets the thesis say something more exact. The tuned point matters, but nearby settings behave similarly enough that the archived full HGB reference family does not look brittle.

3.4.5 Subgroup Review Logic

The subgroup review focuses on q2 and raceeth because the literature and the survey context both make subgroup heterogeneity plausible there. The review uses explicit adequacy thresholds for minimum subgroup size, minimum positives, and minimum negatives. This prevents thin cells from being over-interpreted and keeps the subgroup analysis aligned with the thesis's non-overclaim posture.

The subgroup review is intentionally diagnostic. It asks whether the archived uncalibrated full HGB reference run exposes any obvious subgroup-specific calibration failure or high-error alert under the implemented thresholds. It does not ask whether the model is fair in a broad normative sense. That narrower framing is one of the strengths of the thesis. It performs a real subgroup review without pretending that a thesis-scale project can settle a much larger fairness question.

3.4.6 Interpretation-Stability Logic

Interpretability is handled through permutation-importance stability for the archived full HGB reference model. The central question is whether the bullying indicators remain meaningfully ranked and stably present across folds. If they do, the thesis can report that the bullying block remained engaged in the full HGB reference model. If they do not, then even a positive held-out ablation result would be harder to interpret as a stable model feature.

This is a review of signal engagement, not a causal explanation. The thesis never uses permutation importance to claim that one variable is a mechanistic driver of distress. It uses the stability of the importance pattern to check whether the final block-level story is consistent with the fitted model's behavior across folds. That is a disciplined role for interpretability in a public-health prediction study.

3.4.7 Governance, Misuse, and Non-Deployment Boundaries

Governance is part of the scientific method of this thesis, not an ethical afterthought added at the end. The project uses public-use, self-reported, cross-sectional data about adolescents, so the manuscript needs explicit boundaries on what the model can and cannot justify.

Making these points explicit strengthens the document because it aligns the rhetorical force of the thesis with the real strength of the evidence. A polished thesis that overclaims would still be a weak scientific object. A polished thesis that states its limits precisely is stronger because the reader can trust that its positive claims are not hiding inflation elsewhere.

3.4.8 Documentation Discipline and Traceability

Reproducibility is not confined to back matter. The body itself needs to explain why the analysis is traceable. The thesis accomplishes that in three ways. First, it fixes the split protocol, the feature system, and the model-selection rule. Second, it roots empirical claims in named outputs

such as the analysis-ready sample summary, the full-feature comparison table, the bullying-block ablation table, and the multiseed and bootstrap summaries. Third, it maintains a clear claim-to-evidence chain that is summarized later in the chapter where the argument depends on it.

This reproducibility discipline strengthens the thesis because it makes the analysis traceable from design decisions to reported results. The document is not only an interpretation of findings. It is also a structured account of how those findings were produced and checked.

3.4.9 Subgroup Adequacy Thresholds and Flag Criteria

The subgroup review is only defensible because it uses explicit adequacy thresholds. The thesis does not allow every coded slice to carry equal interpretive weight. Cells that fail the minimum size or event-count thresholds remain reportable, but they are not overread as though they were equally stable. That threshold logic gives the subgroup analysis real methodological discipline.

The thresholds also matter for the structure of the thesis. They prevent the subgroup section from drifting into a generic fairness discussion detached from the actual data density of the study. Instead, the thesis can say something narrower and stronger. It performed a defined subgroup review under explicit adequacy rules and did not find an obvious flagged failure in the eligible cells.

Table 7A. *Subgroup Adequacy Thresholds and Interpretive Implications*

Threshold or trigger	Value used in the project	Interpretation rule
Minimum subgroup size	200	Cells below this are reported but not strongly interpreted
Minimum subgroup positives	20	Supports stable slope interpretation
Minimum subgroup negatives	20	Supports stable slope interpretation
Probability bins	10	Supports local calibration review without overfragmenting the held-out sample
Trigger fields	`low_slope_flag`, `high_error_flag`	Flags potential subgroup-specific instability for explicit review

3.4.10 Traceability of Core Empirical Claims

The body also benefits from a compact documentation table because it shows that the thesis's empirical statements are anchored in identified outputs rather than in memory or prose summary. This is especially important in a thesis that asks the reader to trust a fairly intricate combination of held-out comparison, calibration logic, stability checks, and governance boundaries. A visible claim map makes that complexity more manageable.

Table 7B. Core Empirical Claim Map

Body claim family	Core supporting evidence	Why it matters in the body
Sample size and modeling boundary	Modeling table and eligibility summary	Anchors the analytic sample and outcome counts
Missingness and preprocessing rationale	Missingness summary	Supports the low-complexity preprocessing argument
Final candidate comparison	Final candidate comparison table	Anchors the held-out model-selection chapter
Bullying-block increment	Bullying-block ablation table	Anchors the main research answer
Calibration sensitivity	Calibration sensitivity table	Anchors the probability-quality argument
Robustness and subgroup review	Multiseed, bootstrap, subgroup, and importance summaries	Anchors the bounded-confidence interpretation

3.4.11 Why the Robustness Layers Are Not Interchangeable

The robustness layers in this section may look similar at first glance because all of them qualify the final empirical claim. They are not interchangeable. Seed sensitivity asks whether the archived training process is fragile. Bootstrap intervals ask how much the archived held-out summaries move under resampling. Hyperparameter sensitivity asks whether the full HGB reference configuration rests on a narrow tuning point. Subgroup review asks whether the aggregate story hides obvious failures in reviewed reference-run slices. Interpretation stability asks whether the bullying block remains engaged in the full HGB reference model. The thesis gains evidentiary clarity by not asking any one robustness layer to do the work of all the others.

This distinction also improves the reading experience of the thesis. The section becomes more than a pile of robustness outputs. It becomes a mapped set of methodological controls, each tied to a different possible weakness. That clarity is one of the reasons the final draft reads more like a research thesis than like an overgrown report.

3.4.12 What These Checks Can and Cannot Establish

The layered robustness design buys bounded confidence, not certainty. Multiseed summaries show that the archived uncalibrated full HGB reference configuration is not obviously unstable to nearby training perturbations. Bootstrap intervals show that the corresponding held-out metrics are not implausibly brittle within the observed sample. Hyperparameter sensitivity shows that the full HGB reference family does not owe its behavior to a razor-thin setting choice. Subgroup review shows that the aggregate result is not masking an obvious flagged failure in the evaluated reference-run subgroup cells. Permutation stability shows that the bullying block

remains substantively engaged in the fitted full HGB model. None of these checks establishes external validity, clinical utility, or mechanism. Together, however, they materially strengthen the thesis's internal credibility.

What these analyses do not buy is external validation, clinical relevance, or causal clarity. The thesis is strongest when it says both things at once. The results are more defensible because the study shows what these checks can and cannot establish.

3.4.13 Section Summary

This section gives the final results a stronger foundation. It shows what kinds of additional evidence the thesis has, how those checks differ from one another, and why governance belongs inside the methodological design. The next section can therefore report its empirical findings against a richer scaffold of validation, robustness, and restraint than the earlier draft provided.

4. Results

4.1 Descriptive and Sample Context for the Predictive Task

The results chapter begins with descriptive and sample anchors so the predictive task remains legible as a public-health question before candidate comparisons begin. This section establishes the weighted prevalence context, the analysis-ready sample size, and the stable denominator that all later held-out comparisons share.

4.1.1 Weighted and Raw Descriptive Anchors

The results section begins with descriptive anchors because the predictive problem should remain legible as a public health problem. In the weighted descriptive outputs, QN26 persistent sadness or hopelessness has prevalence 39.6 percent. QN24 bullied on school property has prevalence 17.1 percent. QN25 electronically bullied has prevalence 13.0 percent. In the analysis-ready modeling table, the unweighted QN26 rate is 40.8 percent. These values confirm that the target is common enough for meaningful probability scoring and that the bullying indicators are not so rare that their contribution would be difficult to estimate in a stable held-out design.

The analytic sample counts reinforce this point. The final modeling table contains 19,863 observations, with 8,108 positive outcomes and 11,755 negative outcomes. This is not a tiny convenience sample. It is large enough to support a model-comparison section that is evidence-dense without needing speculative extrapolation. At the same time, the survey remains cross-sectional and public-use, so the results still need to be interpreted within the boundaries established earlier.

Table 8. *Weighted Descriptive Anchors and Modeling-Table Counts*

Quantity	Value	Interpretation
Weighted prevalence of QN26	0.396122	Public health motivation for the target outcome
Weighted prevalence of QN24	0.171142	Public health prevalence of school-property bullying
Weighted prevalence of QN25	0.129854	Public health prevalence of electronic bullying
Final modeling rows	19863	Shared analytic sample for all held-out comparisons
Positive outcome count	8108	Modeling-table count of `y_qn26 = 1`
Negative outcome count	11755	Modeling-table count of `y_qn26 = 0`

Quantity	Value	Interpretation
Unweighted outcome rate	0.408196	Outcome burden inside the analysis-ready table

4.2 Final Candidate Comparison and Model Selection

Before the final full-feature comparison, the project already had evidence that calibration behavior would matter. The baseline-family archive showed that logistic baseline and HGB baseline were almost indistinguishable on ROC AUC while differing meaningfully on Brier score and calibration behavior. That earlier comparison did not decide the final thesis result, but it foreshadowed the exact pattern that later reappeared in the full-feature candidate set. This developmental context belongs in the body because it helps the reader see that calibration-sensitive model selection was not invented after the final table was already known.

It also clarifies why the final HGB versus logistic comparison should not be interpreted as a purely nonlinear victory story. The project had already learned that probability-quality differences could matter more than dramatic discrimination gaps. The full-feature results then confirmed that same methodological lesson under the thesis's main feature system.

Figure 4. Baseline-family calibration comparison preserved from the earlier model-selection stage. The figure shows that probability-quality differences were already visible before the final full-feature comparison was locked in.

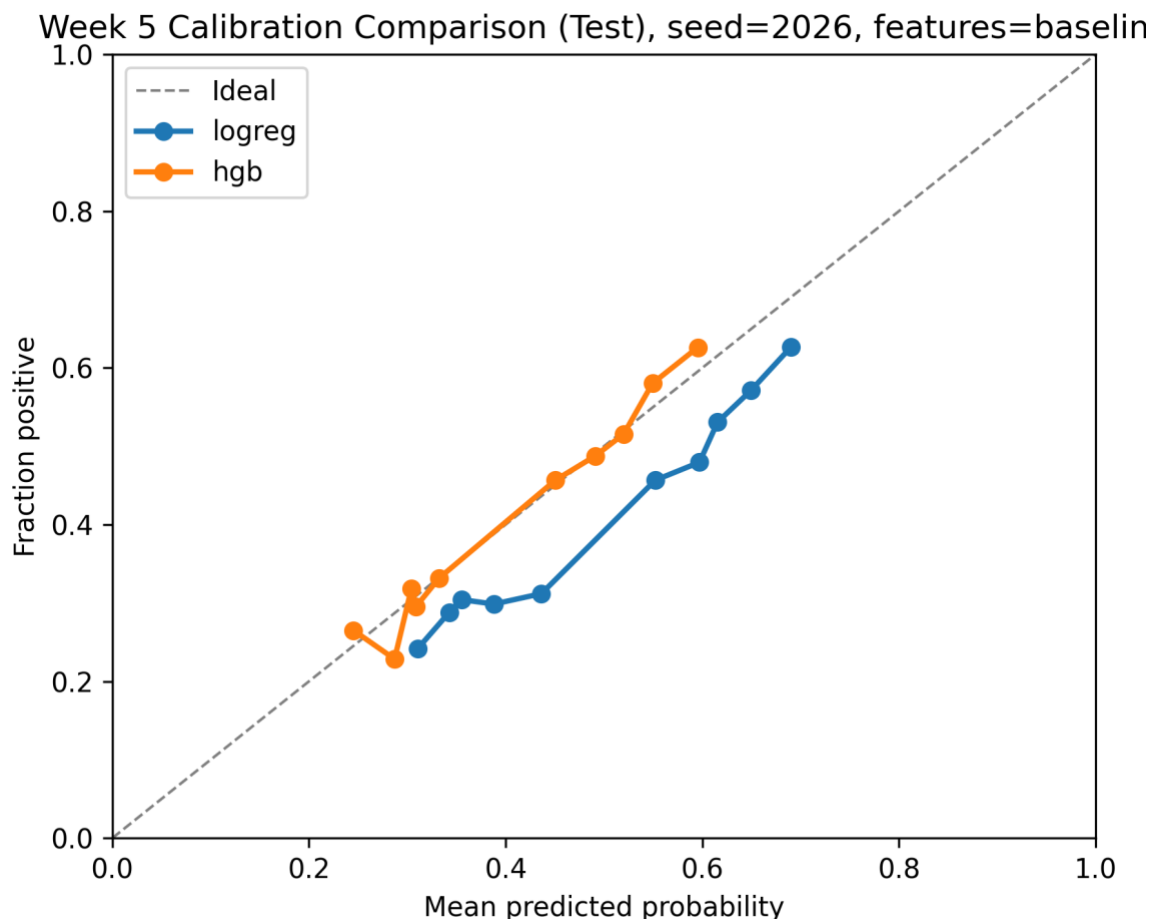


Table 8A. Baseline-Family Calibration Context

Model family	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
Logistic baseline	0.649822	0.547476	0.231985	0.987174	-0.363811
HGB baseline	0.650200	0.537872	0.225224	1.071831	0.036123

4.2.1 Candidate Comparison Under the Full Feature Set

The first core empirical task is to compare the full-feature candidates. Figure 5 and Table 9 show the held-out comparison among logistic regression without post hoc calibration, HGB without post hoc calibration, HGB with Platt scaling, and HGB with isotonic calibration. The most important observation is not that one candidate runs away from the others on discrimination. It is

that the discrimination story is relatively tight while the probability-quality story creates real separation. That is exactly the kind of pattern the thesis design was built to detect.

The comparison also reveals why the thesis needed an explicit model-selection rule. If one focused only on ROC AUC and PR AUC, the Platt-calibrated and uncalibrated HGB candidates would look almost identical and the logistic full model would still appear broadly competitive. Once Brier score and calibration slope are treated as central, the differences become substantively important. That is why the thesis spends so much earlier space on evaluation philosophy. The final candidate table is more informative when the reader understands what the table is being asked to prove.

Figure 5. Held-out candidate selection panel for the final full-feature comparison. Brier score and absolute calibration-slope deviation from one create the decisive separation, while ROC AUC and PR AUC remain comparatively tight.

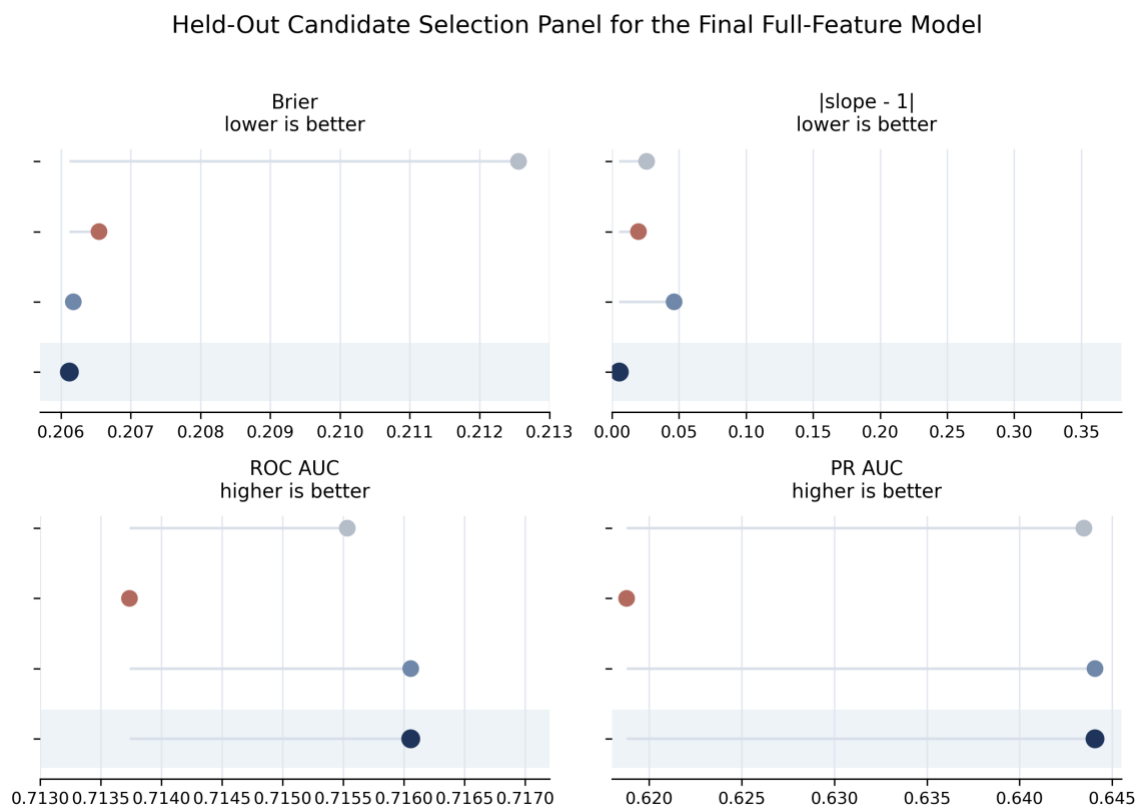


Table 9. Held-Out Full-Feature Candidate Comparison

Candidate model	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
HGB full with Platt scaling	0.716057	0.644073	0.206122	1.005455	0.014371

Candidate model	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
HGB full without post hoc calibration	0.716057	0.644073	0.206178	1.046301	0.027978
HGB full with isotonic calibration	0.713738	0.618789	0.206546	0.980255	-0.000656
Logistic full without post hoc calibration	0.715534	0.643474	0.212558	0.974171	-0.368623

4.2.2 Cross-Validation Alignment with the Held-Out Selection Story

The held-out results are the thesis claim, but it remains useful to know whether the training-side cross-validation comparison told the same story. It largely did. The HGB family already had the best cross-validation Brier profile, and Platt scaling already improved that profile slightly while moving the calibration slope toward one. This alignment does not strengthen the claim beyond the held-out test set, but it does show that the final held-out result was not disconnected from the training-side comparison logic.

That matters because it makes the final selection look more coherent as a model family choice. The thesis is not presenting a model that looked weak in cross-validation and somehow won only on the held-out split. The selected family behaved consistently across the internal and final evaluation layers that the design intentionally separates.

Table 9A. *Cross-Validation Alignment for the Selected Candidate Family*

Candidate row	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
Logistic full cross-validation mean	0.717926	0.637240	0.212722	0.986029	-0.371567
HGB full cross-validation mean	0.716004	0.635046	0.206851	1.042969	0.014692
HGB full with Platt cross-validation mean	0.716004	0.635046	0.206818	0.984161	-0.005695
HGB full with isotonic cross-validation mean	0.714986	0.619679	0.206967	0.949113	-0.022121

4.2.3 Final Model Selection and Why Platt Wins

The selected model is HGB full with Platt scaling. This choice follows directly from the lexicographic rule rather than from any narrative preference for nonlinear models. Relative to the uncalibrated HGB full candidate, Platt scaling leaves ROC AUC and PR AUC unchanged on the held-out test set while improving Brier score from 0.206178 to 0.206122 and moving the calibration slope from 1.046301 to 1.005455. That slope movement is not cosmetic. It moves the final probability surface materially closer to the ideal calibration target.

The comparison with logistic full is equally revealing. Logistic full remains a competent ranking model with held-out ROC AUC 0.715534 and PR AUC 0.643474. Yet its Brier score is materially worse at 0.212558 and its calibration intercept is substantially more negative at -0.368623. This means the logistic reference is informative precisely because it demonstrates that a close ranking result does not imply comparable probability quality. The thesis benefits from reporting that nuance rather than simply celebrating the nonlinear winner.

4.3 Incremental Predictive Value of the Bullying Block

The second core empirical task is the one the thesis was designed around from the beginning. Does the bullying block add predictive value beyond the narrow baseline? Table 10 and Figure 6 answer yes. When the bullying block is added to the non-bullying comparator, held-out ROC AUC increases from 0.650200 to 0.716057. Held-out PR AUC increases from 0.537872 to 0.644073. Held-out Brier score decreases from 0.225224 to 0.206178. Calibration slope moves from 1.071831 to 1.046301 and calibration intercept moves from 0.036123 to 0.027978.

The key point is that the improvement is visible across both discrimination and probability-quality dimensions. A positive AUC delta without a Brier gain would have supported a much weaker conclusion. Here, the bullying block improves the metrics that matter most to the thesis's design. That makes the incremental-value claim stronger than a simple statement that bullied students have worse average outcomes. It shows that the block changes held-out predictive performance in a measurable way.

Figure 6. Held-out and cross-validation deltas when the bullying predictor block is added to the non-bullying comparator. Positive deltas for ROC AUC and PR AUC and negative deltas for Brier indicate better performance with the bullying block.

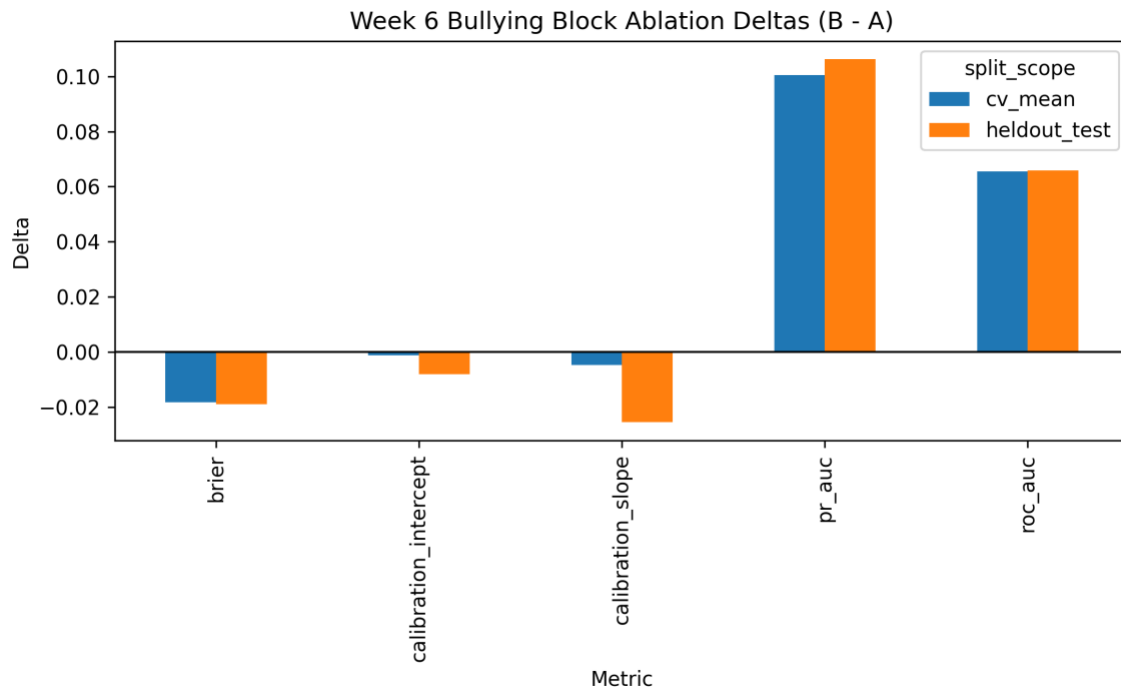


Table 10. Held-Out Incremental Predictive Value of the Bullying Block

Metric	Comparator without bullying	Expanded model with bullying	Delta
ROC AUC	0.650200	0.716057	0.065857
PR AUC	0.537872	0.644073	0.106201
Brier	0.225224	0.206178	-0.019046
Calibration slope	1.071831	1.046301	-0.025529
Calibration intercept	0.036123	0.027978	-0.008146

4.4 Calibration Sensitivity and Probability-Bin Interpretation

The calibration sensitivity results clarify why Platt scaling is the preferred post hoc calibration state for the selected HGB family. Figure 7 and Table 11 show that Platt scaling improves Brier score and calibration slope relative to the uncalibrated HGB full candidate while preserving discrimination. Isotonic calibration, by contrast, lowers ROC AUC and PR AUC and produces a worse Brier score than Platt. Its calibration slope is below one and its intercept is closer to zero

than the uncalibrated HGB family, but the overall tradeoff is weaker once the lexicographic rule is applied.

This is one of the most important places where the thesis surpasses a conventional long paper. The results section does not simply list calibration numbers. It shows how calibration actually changes the final model-choice narrative. Without this section, the final selection would look arbitrary or overly technical. With it, the reader can see that probability quality is not an abstract preference. It is the specific reason one candidate becomes the final model.

Figure 7. Calibration sensitivity across no post hoc calibration, Platt scaling, and isotonic regression for the selected HGB full model family. Platt scaling produced the best balance of Brier score and calibration slope.

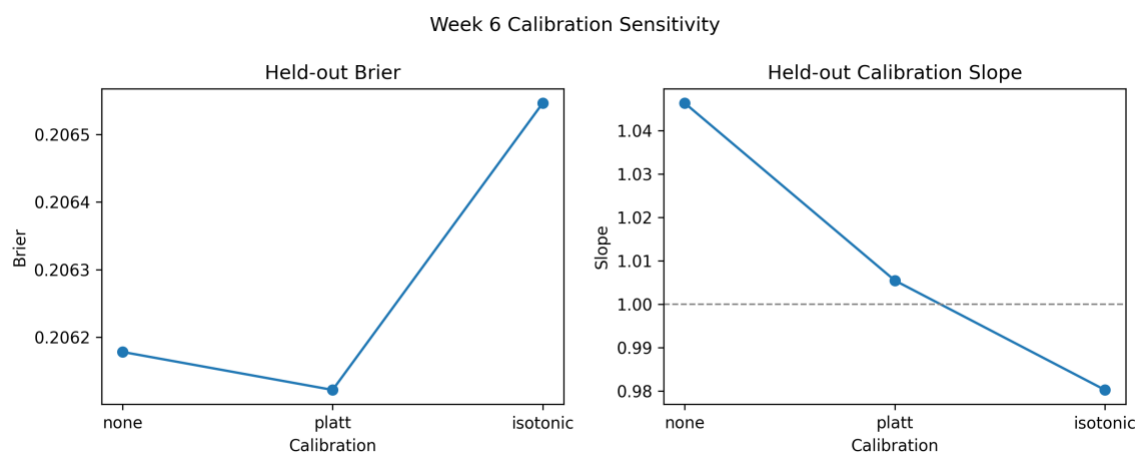


Table 11. Held-Out Calibration Sensitivity for the Selected HGB Family

Calibration method	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
None	0.716057	0.644073	0.206178	1.046301	0.027978
Platt	0.716057	0.644073	0.206122	1.005455	0.014371
Isotonic	0.713738	0.618789	0.206546	0.980255	-0.000656

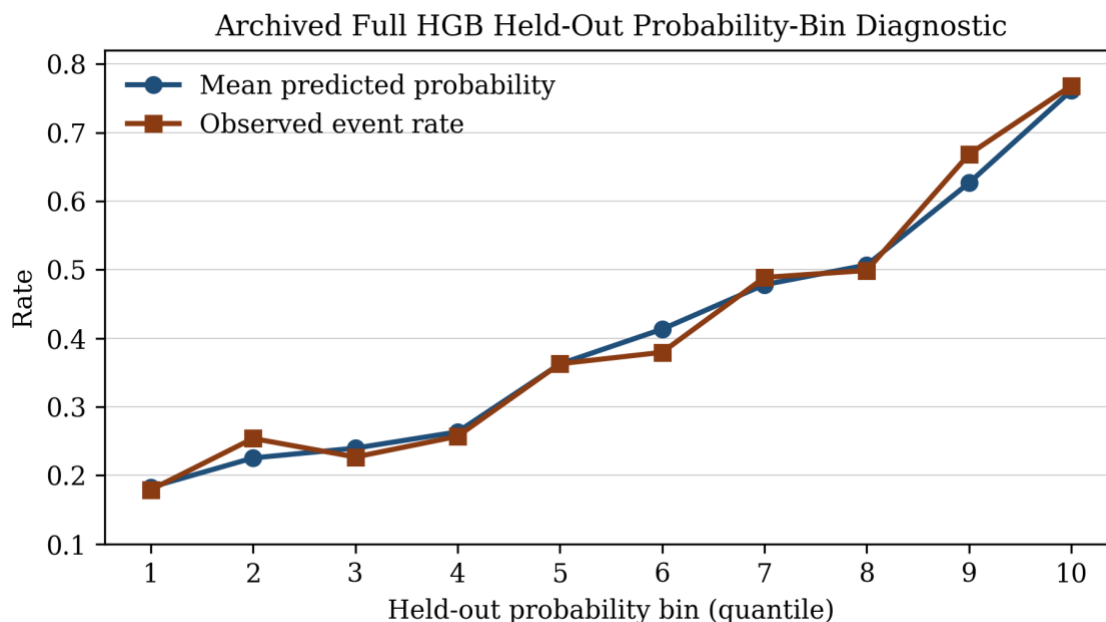
4.4.1 Probability-Bin Diagnostic Interpretation

The proposal required error analysis by predicted-probability bins, and the retained held-out prediction stream is sufficient to support that analysis. Figure 8 provides the thesis-level summary of the bin diagnostic using the archived held-out HGB full prediction stream without post hoc calibration. The most visible gaps appear in the second, sixth, and ninth bins, where observed-minus-predicted differences are especially notable.

This figure matters because it turns the calibration discussion into something concrete. The slope and intercept summaries already establish that probability quality differs across calibration states.

The bin diagnostic shows how those differences manifest across the score range. It is therefore a bridge between formal metric reporting and intuitive error interpretation.

Figure 8. Probability-bin diagnostic for the archived held-out full HGB prediction stream without post hoc calibration. Positive values indicate underprediction within a bin, while negative values indicate overprediction.



4.5 Robustness and Subgroup Review

The final result depends on more than one held-out table. The archived multiseed and bootstrap summaries do not rerun the final Platt-selected candidate. Instead, they audit the adjacent uncalibrated full HGB reference configuration that underlies the family-level stability checks. Across seeds 2026 through 2029, mean cross-validation ROC AUC ranges only from 0.715906 to 0.716817. Mean PR AUC ranges from 0.633901 to 0.636567. Mean Brier ranges from 0.206746 to 0.206851. Those ranges are extremely tight relative to the scale of the thesis question.

The bootstrap layer supports a related conclusion from a different angle. For the archived uncalibrated full HGB reference run, the held-out ROC AUC 95 percent interval spans 0.698754 to 0.733358. The held-out Brier interval spans 0.201062 to 0.211181. The held-out calibration slope interval spans 0.956962 to 1.146544. These ranges show that the archived point estimates are not implausibly sharp, but they also do not suggest a model so unstable that the broader narrative depends on one lucky held-out draw.

Figure 9. Multiseed stability summary for the archived uncalibrated full HGB reference configuration. Cross-validation summaries remain tightly clustered across nearby seed perturbations.

Multiseed Stability for the Selected Full HGB Family

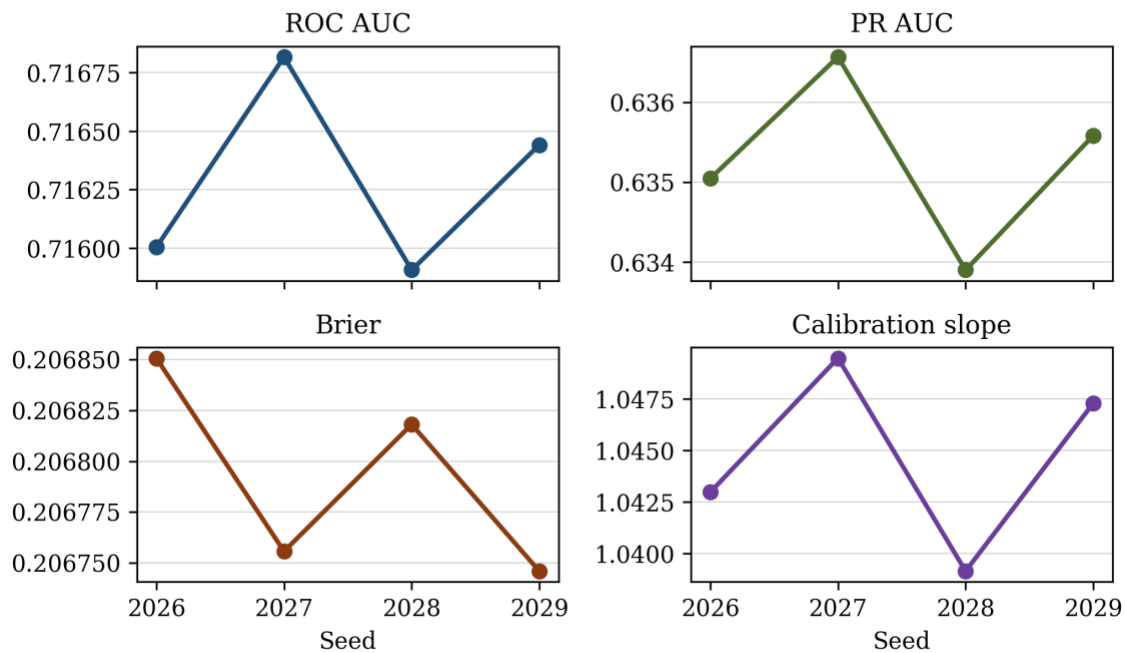


Figure 10. Held-out bootstrap interval summary for the archived uncalibrated full HGB reference configuration. Interval width is visible for the archived reference metrics used in the family-level stability audit.

Held-Out Bootstrap Interval Summary for the Selected Model

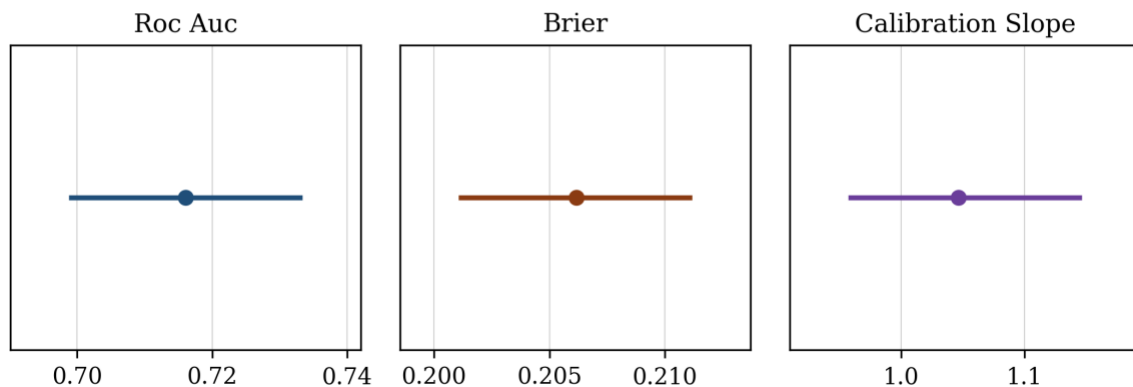


Table 12. *Stability and Uncertainty Synthesis*

Evidence dimension	Numeric summary	Interpretation
Multiseed cross-validation stability	ROC AUC means 0.715906 to 0.716817, PR AUC means 0.633901 to 0.636567, Brier means 0.206746 to 0.206851	Selected family remains highly stable across nearby seed perturbations
Held-out bootstrap intervals	ROC AUC 0.698754 to 0.733358, Brier 0.201062 to 0.211181, calibration slope 0.956962 to 1.146544	Main held-out summaries remain within a reasonably tight uncertainty band
Hyperparameter sensitivity	Cross-validation Brier remains within 0.206786 to 0.206990 across local perturbations	Final HGB family does not sit on an obviously unstable local configuration

4.5.1 Subgroup Review and Interpretation Stability

The subgroup and permutation-importance results supply another layer of evidence. In the q2 subgroup review of the archived uncalibrated full HGB reference run, the two defined groups have ROC AUC values of 0.681 and 0.645, Brier scores of 0.223 and 0.190, and calibration slopes of 1.173 and 0.978. In the race or ethnicity audit, defined groups have ROC AUC ranging from 0.677 to 0.768, Brier ranging from 0.176 to 0.223, and calibration slope ranging from 0.872 to 1.435, again without triggered flags among the adequately documented cells. The absence of triggered flags is reassuring. It is not a fairness guarantee.

Permutation-importance stability sharpens the block-level interpretation further. q2 remains the strongest feature by mean negative-Brier importance at 0.022014. Both bullying indicators remain immediately behind it, with x_{qn25} at 0.014915 and x_{qn24} at 0.013076. Both bullying indicators are stable across all folds. This is exactly the pattern the thesis hoped to see. The bullying block matters in the archived full HGB reference model, but it does so alongside rather than instead of the strongest demographic signal.

Table 13. Subgroup review and permutation-importance stability summary for the archived full HGB reference model.

Audit component	Verified summary	Interpretation boundary
Sex subgroup review	ROC AUC 0.681 and 0.645, Brier 0.223 and 0.190, calibration slope 1.173 and 0.978, no triggered flags	No obvious sex-specific failure under the implemented thresholds
Race or ethnicity subgroup review	ROC AUC 0.677 to 0.768, Brier 0.176 to 0.223, calibration slope 0.872 to	Supports cautious reporting, not a strong fairness claim

Audit component	Verified summary	Interpretation boundary
	1.435, no triggered flags in well-documented cells	
Permutation-importance stability	`q2` rank 1, `x_qn25` rank 2, `x_qn24` rank 3, `raceeth` rank 4, both bullying variables stable across all folds	Bullying block remains materially engaged in the selected model

4.5.2 End-of-Results Synthesis

The results section supports a restrained but positive answer to the thesis question. The bullying block adds measurable predictive value beyond the narrow baseline. Probability quality, not just ranking, improves. The selected HGB family remains the strongest candidate under the thesis's stated selection rule, and the surrounding diagnostic evidence makes that result more legible and more bounded.

Just as important, each evidence layer has a distinct job. Descriptive anchors explain why the problem matters. The full-feature candidate comparison identifies the strongest family under the stated rule. The bullying-block ablation isolates the main research question. Calibration sensitivity explains why the final model is selected. The bin diagnostic interprets what calibration means across the score range. Robustness and subgroup review show that the main result is not obviously fragile. Interpretation stability shows that the bullying block remains engaged in the selected family. The results section therefore reads as a thesis section because its components are cumulative rather than repetitive.

Expanded diagnostic detail is preserved in Appendix A, and the threshold-oriented robustness extensions are preserved in Appendix B, so the main results chapter can remain focused on the central held-out argument, the calibration-centered selection logic, and the bounded robustness story that directly answer the thesis question.

5. Discussion

5.1 Interpretation of Main Findings

This thesis asked whether two bullying exposure indicators from the 2023 National Youth Risk Behavior Survey add incremental predictive value for persistent sadness or hopelessness beyond a narrow demographic baseline under a fixed held-out protocol. Within that design, the answer is yes. The final selected model, histogram gradient boosting with Platt scaling, achieved held-out ROC AUC of 0.716, PR AUC of 0.644, Brier score of 0.206, calibration slope of 1.005, and calibration intercept of 0.014. Relative to the non-bullying comparator, adding QN24 and QN25 improved ROC AUC by 0.066, PR AUC by 0.106, and Brier score by 0.019. Those gains are not cosmetic. They indicate that the bullying block changes both rank ordering and probability quality in a way that is detectable after demographic adjustment.

The main substantive interpretation is therefore narrower and stronger than a generic claim that bullying matters. Prior literature already shows that bullying victimization is associated with adolescent mental health burden, including persistent sadness, suicidality, and related harms. The contribution here is that two YRBS bullying indicators remain useful as out-of-sample predictors after a disciplined baseline is locked. That finding is consistent with the broader literature summarized by Ye et al. (2023), Young et al. (2024), Kreski et al. (2022), English et al. (2024), and Agarwal et al. (2026), but it answers a different question. It addresses predictive increment, not etiologic mechanism.

It also matters that the thesis kept school-property bullying and electronic bullying distinct rather than collapsing them into one blended exposure. That decision was supported by the literature and justified methodologically. Cyberbullying is not simply a trivial online extension of offline bullying, and the two indicators can plausibly capture different exposure settings and reporting patterns. The observed incremental gain therefore supports the design choice to treat the bullying block as substantively motivated and internally differentiated, while still evaluating it through one fixed predictive comparison rather than through coefficient storytelling.

The results also clarify what a responsible interpretation looks like. Better held-out prediction does not establish that bullying causes persistent sadness or hopelessness in the causal sense. The YRBS is a cross-sectional surveillance instrument, not a causal identification design. The thesis can therefore say that adding the bullying block improves prediction for QN26 in this frozen protocol. It cannot say that modifying bullying exposure by itself would yield the same numerical change in risk. Keeping that distinction explicit makes the positive result more credible rather than less valuable.

5.2 Relation to Prior Literature and Methodological Contribution

The main results close the literature gap identified earlier in the thesis. Prior YRBS-based and bullying-focused studies largely establish association, prevalence, or disparity patterns, whereas the present study asks whether a compact bullying block adds held-out predictive value under calibration-aware evaluation. The strongest methodological contribution of the thesis is not merely that one model family won. It is that the manuscript treats probability quality as a first-class evaluation target and organizes the evidence accordingly. The revised structure makes that contribution clearer by separating the inferential spine of the paper from the broader diagnostic archive, so that calibration, model selection, and the bullying-block comparison remain visible as the core analytic sequence rather than being submerged inside tables.

That design choice explains why the final model is not described as the best model simply because it posted the highest ROC AUC. The selected model won because it combined acceptable discrimination with the strongest overall probability profile under the predeclared rule. The final held-out slope near 1.0 and intercept near 0, together with the best reported Brier score among the final candidates, support the claim that the output probabilities are meaningfully better behaved than the closest alternatives. Platt scaling matters here because it improved the calibration profile without forcing the thesis to abandon the original comparison structure.

A second contribution is the explicit weighting boundary. Weighted descriptive summaries were used to preserve population-facing prevalence context, but the primary predictive comparison was intentionally left on an unweighted held-out benchmark. That separation prevented the paper from blending two analytic tasks into one unstable claim. In the thesis body, weighting serves descriptive surveillance interpretation. In the predictive benchmark, the core question is whether the bullying block improves model performance under one fixed protocol. Keeping those roles separate makes the empirical contrast easier to interpret and easier to reproduce.

The credibility of the result also depends on the supporting layers that were treated as distinct from primary model selection rather than as decorative extras. Multiseed checks, bootstrap summaries, subgroup review, and interpretation-stability analysis do not prove universal validity. They do something more limited and more useful. They show that the main result is not obviously dependent on one lucky split, one fragile importance pattern, or one aggregate metric that hides subgroup instability in the adjacent archived HGB diagnostics. In a thesis of this scale, that layered structure is a methodological strength because it turns the final claim into a documented sequence rather than a single leaderboard snapshot.

Relatedly, the manuscript treats reproducibility as part of the scientific argument rather than as an implementation appendix. Frozen splits, logged design decisions, evidence mapping, and traceable reporting boundaries reduce the risk that the narrative outruns the artifacts. That is

especially important in a public-use survey setting where many apparently reasonable analytic variants could have been pursued. The finished thesis is stronger because it makes those forks visible and because it explains why the chosen design remained narrow.

The paper also benefits from being legible as a prediction-model report rather than only as a machine-learning exercise. In practice, that means the thesis states the target, sample restrictions, predictor blocks, preprocessing logic, split discipline, calibration procedure, and selection rule in a form that another reader could audit. That posture is consistent with the transparent development, validation, and reporting expectations emphasized by Efthimiou et al. (2024), and it strengthens the credibility of the empirical claim because it reduces ambiguity about what was actually compared.

5.3 Limitations, Governance, and Appropriate Use

The study has several important limitations. First, the YRBS is a self-report, cross-sectional surveillance dataset. Measures are collected at one time point and are subject to reporting error, recall limitations, and instrument-specific wording constraints. QN26 captures persistent sadness or hopelessness as a surveillance indicator, not a clinical diagnosis. QN24 and QN25 capture reported bullying exposures, not verified events or longitudinal exposure histories. The thesis therefore operates within the measurement authority of the public instrument and should be read accordingly.

Second, the final predictive contrast is intentionally narrow. That is a design strength for internal interpretability, but it also limits how much of the outcome surface the model can explain. The baseline omits many other potential predictors that may matter for adolescent distress, including richer family, school, sleep, substance-use, and broader digital-context variables. The present result should therefore be read as evidence that the bullying block adds value beyond this baseline, not as evidence that the finished model is close to a complete representation of adolescent mental health risk.

Third, external validity remains unresolved. The manuscript uses one national 2023 public-use file under one frozen split. It does not test transport to other years, states, districts, or adjacent survey systems. It also does not establish that calibration quality would remain stable if prevalence, response behavior, or survey composition changed. That caution is consistent with the YRBS documentation and with the broader literature on prediction-model transportability. The current study supports bounded in-sample and held-out conclusions within one dataset. It does not by itself justify deployment, screening, or institutional action.

These limits are exactly why governance belongs in the main discussion. A research-stage model trained on sensitive youth mental-health data should not be framed as a hidden risk detector for

disciplinary targeting, individual diagnosis, or automated triage. The correct use of the final model package is analytical and methodological. It helps quantify whether two bullying indicators carry incremental predictive information for one defined outcome under one documented protocol. Prohibited-use language is part of the claim boundary that prevents predictive association from being mistaken for causal, clinical, or operational authority.

The subgroup results require the same discipline. Aggregate success does not guarantee equal stability or equal calibration everywhere. Subgroup review in this thesis functions as a diagnostic validity layer with adequacy thresholds, not as a final fairness certification. That is the appropriate posture for a study of this size and scope. It acknowledges heterogeneity without pretending that a single national held-out result resolves every subgroup question.

5.4 Future Research

The most defensible next step is not uncontrolled feature expansion. It is extension with protocol continuity. One immediate direction is to test weighting sensitivity more directly in model fitting while preserving the same target, split logic, metric framework, and reporting discipline. Another is to expand the covariate set in a staged way so that additional predictors can be evaluated without blurring the original bullying-block contrast. Both directions would be more informative if they preserved the thesis principle of changing one design dimension at a time.

A second priority is external validation and temporal transport. Replication across later YRBS waves, or across state and district files with comparable coding, would show whether the bullying increment and the calibration profile persist beyond the frozen 2023 national context. That work should also include subgroup-disaggregated calibration analysis and recalibration studies so that future readers can distinguish stable signal from context-specific fit.

That next phase should also treat transport as a calibration problem, not only a discrimination problem. As Huang et al. (2020) and Silva Filho et al. (2023) make clear, probability calibration can deteriorate even when rank ordering remains comparatively stable. Future validation should therefore report those shifts explicitly rather than assuming that a single national 2023 fit will generalize unchanged.

A third priority is translation of probabilities into clearer interpretive communication without violating the research boundary. Threshold-specific summaries, decision-curve style analyses, or structured risk communication studies could all be valuable, but they should occur only after external validation and calibration durability are better established. In this thesis, thresholds are discussed as reporting devices rather than as action triggers. Future work can refine that interpretation layer, but only after the underlying probability engine has been validated more broadly.

Taken together, the study contributes a clear answer to a narrow predictive question and a disciplined template for how such questions should be evaluated in survey-based mental-health research. The thesis shows that a modest, well-specified contrast can produce a meaningful empirical result when calibration, robustness, documentation, and governance are treated as integral parts of the design. That combination is the strongest basis on which the present findings should be carried forward.

6. Conclusion

This thesis asked whether two bullying exposure indicators from the 2023 Youth Risk Behavior Survey add incremental predictive value for persistent sadness or hopelessness after demographic adjustment under a fixed held-out protocol. Within that design, the answer is affirmative. Adding the bullying block improved held-out ROC AUC by 0.066, PR AUC by 0.106, and Brier score by 0.019 relative to the non-bullying comparator. The final selected model, histogram gradient boosting with Platt scaling, achieved ROC AUC of 0.716, PR AUC of 0.644, Brier score of 0.206, calibration slope of 1.005, and calibration intercept of 0.014. Those results support the hypothesis that the bullying indicators contribute measurable predictive information beyond the locked demographic baseline.

The larger contribution is methodological. The thesis demonstrates how a survey-based public-health prediction study can separate population description from held-out prediction, discrimination from calibration, model comparison from threshold interpretation, and research evidence from operational use. That structure matters because the substantive finding is only as credible as the protocol that produced it. By preserving a narrow baseline, a fixed bullying block, a frozen evaluation split, a probability-first selection rule, and explicit governance limits, the manuscript gives the empirical result a clear interpretive boundary.

The findings should therefore be carried forward as bounded predictive evidence, not as causal proof or as a deployment recommendation. The YRBS is cross-sectional, self-reported, and designed for surveillance. The model is useful for quantifying incremental signal in a documented research design. It is not a diagnostic instrument, a screening tool, or a justification for automated decisions about individual students. This distinction does not weaken the result. It defines the conditions under which the result is scientifically defensible.

Future work should extend the design through external validation, later YRBS waves, richer but pre-specified covariate systems, and subgroup-disaggregated calibration studies. The priority should be protocol continuity rather than uncontrolled feature expansion. The present thesis provides a strong baseline for that next work: a clear outcome, a transparent predictor-block contrast, calibrated held-out evaluation, and an explicit account of what the evidence can and cannot support.

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Appendix A. Supplementary Diagnostic Archive

This appendix preserves the broader coding, descriptive, comparison, calibration, and robustness tables that support the main thesis argument but are too numerous for the primary results narrative. All appendix materials remain tied to the documented submission-pack artifacts and do not alter the main inference.

A.1 Variable Architecture, Sample Construction, and Raw Coding Anchors

This section consolidates the variable dictionary, analysis-ready sample summary, focused missingness summary, schema excerpt, sample-construction summary, and coding anchors into the body of the thesis. Bringing these materials into the main text makes the predictive design fully inspectable without sending the reader to supplementary materials for essential context.

The central point is methodological transparency. The thesis question is narrow, and the corresponding variable architecture is intentionally sparse. Tables 14 through 19 document exactly how the final analysis-ready table was assembled, how missingness was handled, and how the raw YRBS response anchors connect to the binary features used in the locked predictive comparison.

Table 14. Analysis-Ready Modeling Variable Dictionary

analysis_name	source_column	role	type	coding	missingness handling	notes
`y_qn26`	QN26	target	binary	Yes coded 1, no coded 0, special missing codes mapped to NA	Drop rows with missing target	Primary target for all held-out comparisons
`x_qn24`	QN24	exposure	binary	Yes coded 1, no coded 0, special missing codes mapped to NA	Retain for simple preprocessing inside the pipeline	School-property bullying indicator
`x_qn25`	QN25	exposure	binary	Yes coded 1, no coded 0, special missing codes mapped to NA	Retain for simple preprocessing inside the pipeline	Electronic bullying indicator

analysis_name	source_column	role	type	coding	missingness handling	notes
`q1`	q1	covariate	categorical	Stored as coded category labels	Retain for simple preprocessing inside the pipeline	Age proxy in the locked baseline
`q2`	q2	covariate	categorical	Stored as coded category labels	Retain for simple preprocessing inside the pipeline	Sex proxy in the locked baseline
`q3`	q3	covariate	categorical	Stored as coded category labels	Retain for simple preprocessing inside the pipeline	Grade proxy in the locked baseline
`raceeth`	raceeth	covariate	categorical	Stored as coded category labels	Retain for simple preprocessing inside the pipeline	Race or ethnicity proxy in the locked baseline
`weight`	weight	design	design	Preserved as supplied	Preserved for descriptive work only	Not used as a predictive feature
`stratum`	stratum	design	design	Preserved as supplied	Preserved for descriptive work only	Not used as a predictive feature
`psu`	psu	design	design	Preserved as supplied	Preserved for descriptive work only	Not used as a predictive feature

Table 15. Analysis-Ready Sample Construction Summary

raw_rows	modeling_rows	modeling_cols	y_qn26_n	y_qn26_n1	y_qn26_n0	y_qn26_rate
20103	19863	14	19863	8108	11755	0.408196

Table 16. Focused Missingness Summary for Final Modeling Variables

column	n_missing	missing_rate
y_qn26	0	0.0000
x_qn24	156	0.0079
x_qn25	146	0.0073

column	n_missing	missing_rate
q1	79	0.0040
q2	144	0.0073
q3	163	0.0082
raceeth	351	0.0177

Table 17. Schema Excerpt for Baseline and Design Fields

column	dtype_inferred	n_unique	example values	notes
raceeth	floating	8	`["5", "8", "3", "7", "6"]`	coded public-use field
q1	floating	7	`["3", "4", "5", "6", "7"]`	coded public-use field
q2	floating	2	`["1", "2"]`	coded public-use field
q3	floating	5	`["1", "3", "2", "4", "5"]`	coded public-use field
weight	floating	2699	`["0.8614", "0.892", "0.5081", "1.1759", "0.8453"]`	high-cardinality design field
stratum	integer	16	`["103", "300", "204", "214", "212"]`	design field
psu	integer	82	`["16294", "58418", "58675", "58391", "87717"]`	design field

Table 18. Sample-Construction Summary

Sample-construction step	Rows	Interpretive note
Raw public-use source rows	20103	Starting point before thesis-specific outcome filtering
Rows removed for missing primary outcome	240	Outcome-based exclusion only
Final analysis-ready modeling rows	19863	Shared sample for all locked predictive comparisons

Table 19. Raw-Source Value Counts for the Core Thesis Items

Item	Raw value	Count	Proportion	Thesis-facing meaning
QN24	1	4046	0.201263	Yes, bullied on school property
QN24	2	15856	0.788738	No, not bullied on school property
QN24	'<NA>'	201	0.009999	Missing
QN25	1	3423	0.170273	Yes, electronically bullied
QN25	2	16475	0.819529	No, not electronically bullied
QN25	'<NA>'	205	0.010197	Missing
QN26	1	8108	0.403323	Yes, persistent sadness or hopelessness
QN26	2	11755	0.584739	No, persistent sadness or hopelessness
QN26	'<NA>'	240	0.011939	Missing

A.2 Weighted Heterogeneity and Descriptive Distribution Context

The predictive claim in this thesis is unweighted and held-out, but the public-health context still benefits from a fuller descriptive accounting. The next set of tables preserves the weighted national anchor, weighted heterogeneity across age and race or ethnicity proxies, and the observed response distributions for the core bullying and outcome items.

These tables do not change the model-selection result. Their role is interpretive. They show the prevalence environment in which the predictive task sits and make clear why subgroup review later in the manuscript is not decorative. Even in a tightly bounded predictive study, the national descriptive frame remains part of responsible interpretation.

Table 20. Proposal-Reported National Trend Context Summary

Indicator	Earliest year noted	Most recent year noted	Proposal-reported descriptive pattern
QN26 persistent sadness or hopelessness	1999	2023	28.27 percent in 1999, 39.72 percent in 2023, peak 42.35 percent in 2021

Indicator	Earliest year noted	Most recent year noted	Proposal-reported descriptive pattern
QN29 attempted suicide	1991	2023	7.27 percent in 1991, 9.46 percent in 2023
QN24 bullied on school property	2009	2023	Included for descriptive availability context only
QN25 electronically bullied	2011	2023	Included for descriptive availability context only

Table 21. 2023 Weighted Descriptive Anchor for the Core Thesis Indicators

Measure	2023 weighted prevalence	Role in thesis
QN26 persistent sadness or hopelessness	39.6%	Primary modeled outcome
QN24 bullied on school property	17.1%	First bullying indicator
QN25 electronically bullied	13.0%	Second bullying indicator
QN29 attempted suicide	7.9%	Descriptive reference point only

Table 22. Weighted QN26 Prevalence by Age Proxy

group_value	n	weighted_n	y_qn26_weighted_rate	y_qn26_ci95_low_approx	y_qn26_ci95_high_approx
cat_3	238	518.6752	0.381065	0.339132	0.422998
cat_4	527	1115.5844	0.392061	0.363369	0.420754
cat_5	511	1040.2444	0.428502	0.398380	0.458623
cat_6	504	1070.3769	0.386237	0.357023	0.415452
cat_7	211	404.6613	0.380410	0.332907	0.427913

Table 23. Weighted QN26 Prevalence by Race or Ethnicity Proxy

group_value	n	weighted_n	y_qn26_weighted_rate	y_qn26_ci95_low_approx	y_qn26_ci95_high_approx
cat_1	122	9.9584	0.353521	-0.008056	0.715097
cat_2	73	120.3027	0.321102	0.236462	0.405741
cat_3	218	557.7666	0.390834	0.350216	0.431452

group_value	n	weighted_n	y_qn26_weighted_rate	y_qn26_ci95_low_approx	y_qn26_ci95_high_approx
cat_5	536	1117.9194	0.420876	0.391891	0.449861
cat_6	253	620.5097	0.408307	0.369527	0.447088
cat_7	537	1283.4357	0.383050	0.356419	0.409681

Table 24. Observed response distribution for the primary outcome item QN26.

value	count	proportion
2.0	11755	0.5847
1.0	8108	0.4033
	240	0.0119

Table 25. Observed response distribution for the school-property bullying item QN24.

value	count	proportion
2.0	15856	0.7887
1.0	4046	0.2013
	201	0.01

Table 26. Observed response distribution for the electronic bullying item QN25.

value	count	proportion
2.0	16475	0.8195
1.0	3423	0.1703
	205	0.0102

Table 27. Weighted overall prevalence estimates for the primary outcome and related mental-health indicators.

variable	weighted_n	weighted_rate	ci95_low_approx	ci95_high_approx
y_qn26	4170.7418	0.3961	0.3813	0.411
x_qn24	4022.5919	0.1711	0.1595	0.1828
x_qn25	4104.4513	0.1299	0.1196	0.1401
y_qn27	3844.389	0.1928	0.1803	0.2052
y_qn28	4036.1829	0.1415	0.1307	0.1522
y_qn29	4114.1702	0.0792	0.071	0.0875

variable	weighted_n	weighted_rate	ci95_low_approx	ci95_high_approx
y_qn30	4135.1141	0.0137	0.0102	0.0173

A.3 Metric Definitions, Calibration Roles, and Probability-Bin Interpretation

The main body already states the evaluation philosophy, but the diagnostic section can formalize the interpretive quantities more explicitly. The following tables preserve the metric definitions, calibration-method roles, probability-bin diagnostic values, and the guide used to translate local bin-level gaps into a readable interpretation.

This material matters because the thesis is not a leaderboard exercise. The logic of the project depends on understanding what each metric can and cannot establish. Probability quality, calibration compression or expansion, and local bin-level deviations are separate diagnostic quantities, and the thesis treats them as such.

Table 28. Metric Definitions and Interpretive Roles

Quantity	Definition used in the thesis	Interpretive role
ROC AUC	Probability that a randomly selected positive case receives a higher score than a randomly selected negative case	Threshold-free discrimination summary
PR AUC	Area under the precision-recall curve over the held-out score ranking	Discrimination summary retained because the outcome is not balanced
Brier score	Mean squared probability error over held-out observations	Primary probability-quality criterion
Calibration intercept	Intercept from the logistic recalibration model	Average overprediction or underprediction
Calibration slope	Slope from the logistic recalibration model	Probability compression or extremity

Table 29. Calibration Method Roles in the Thesis

Calibration approach	Fitting boundary	Model form	Role in this thesis
None	No post hoc calibrator	Raw model probabilities	Reference calibration state
Platt scaling	Training-derived predictions only	Logistic recalibration of model scores	Selected calibration state for the final HGB family
Isotonic regression	Training-derived predictions only	Monotone nonparametric calibration map	Sensitivity comparator

Table 30. Probability-Bin Diagnostic for Archived Full HGB Held-Out Predictions

Probability bin	n	Mean predicted probability	Observed event rate	Observed minus predicted
1	397	0.182612	0.176322	-0.006290
2	398	0.225306	0.253769	0.028463
3	397	0.239623	0.221662	-0.017961
4	397	0.263305	0.261965	-0.001340
5	397	0.362398	0.362720	0.000322
6	398	0.413198	0.379397	-0.033801
7	397	0.478096	0.491184	0.013088
8	397	0.506759	0.491184	-0.015575
9	398	0.626499	0.668342	0.041843
10	397	0.761361	0.775819	0.014458

Table 31. Probability-Bin Interpretation Guide

Diagnostic quantity	Positive value means	Negative value means	Thesis use
Observed minus predicted within a bin	Underprediction in that score region	Overprediction in that score region	Appendix-level interpretation support
Calibration intercept	Average probabilities are too low	Average probabilities are too high	Global calibration summary
Calibration slope	Predictions are too compressed when greater than one	Predictions are too extreme when less than one	Global calibration summary

A.4 Expanded Candidate, Calibration, and Hyperparameter Comparisons

The results section reports the decisive candidate rows. This section preserves the wider comparison archive so the reader can inspect the broader evidence pattern rather than only the winning summary. The expanded tables include cross-validation and held-out candidate rows, fuller calibration sensitivity views, bullying-block ablation output, and local hyperparameter perturbations around the selected HGB configuration.

A rigorous thesis should show not only which model won but also what the nearby alternatives looked like and why they lost. These tables therefore function as a check against selective reporting. They expose the local comparison neighborhood around the final choice and show that the selected family remained competitive under adjacent but substantively secondary decision views.

Table 32. Expanded Week 6 Full-Feature Comparison

split_scope	model	feature set	calibration_method	roc_auc	pr_auc	brier	calibration_slope	calibration_intercept
cv_mean	logreg	full	none	0.717926	0.637240	0.212722	0.986029	-0.371567
cv_mean	hgb	full	none	0.716004	0.635046	0.206851	1.042969	0.014692
heldout_test	logreg	full	none	0.715534	0.643474	0.212558	0.974171	-0.368623
heldout_test	hgb	full	none	0.716057	0.644073	0.206178	1.046301	0.027978

Table 33. Full Calibration Sensitivity Summary

split_scope	calibration_method	roc_auc	pr_auc	brier	calibration_slope	calibration_intercept	delta_brier_vs_none
cv_mean	none	0.716004	0.635046	0.206851	1.042969	0.014692	0.000000
cv_mean	platt	0.716004	0.635046	0.206818	0.984161	-0.005695	-0.000033
cv_mean	isotonic	0.714986	0.619679	0.206967	0.949113	-0.022121	0.000116
heldout_test	none	0.716057	0.644073	0.206178	1.046301	0.027978	0.000000
heldout_test	platt	0.716057	0.644073	0.206122	1.005455	0.014371	-0.000056
heldout_test	isotonic	0.713738	0.618789	0.206546	0.980255	-0.000656	0.000368

Table 34. Expanded Bullying-Block Ablation Summary

split_scope	metric	model_a_name	model_b_name	model_a_value	model_b_value	delta_b_minus_a
cv_mean	roc_auc	hgb_full_minus_bullying_none	hgb_full_none	0.650538	0.716004	0.065466
cv_mean	pr_auc	hgb_full_minus_bullying_none	hgb_full_none	0.534563	0.635046	0.100483
cv_mean	brier	hgb_full_minus_bullying_none	hgb_full_none	0.225156	0.206851	-0.018305
heldout_test	roc_auc	hgb_full_minus_bullying_none	hgb_full_none	0.650200	0.716057	0.065857

split_sc ope	metric	model_a_name	model_b_n ame	model_a_v alue	model_b_v alue	delta_b_min us_a
heldout_ test	pr_auc	hgb_full_minus_bullyi ng_none	hgb_full_n one	0.537872	0.644073	0.106201
heldout_ test	brier	hgb_full_minus_bullyi ng_none	hgb_full_n one	0.225224	0.206178	-0.019046
heldout_ test	calibration_slo pe	hgb_full_minus_bullyi ng_none	hgb_full_n one	1.071831	1.046301	-0.025529
heldout_ test	calibration_int ercept	hgb_full_minus_bullyi ng_none	hgb_full_n one	0.036123	0.027978	-0.008146

Table 35. HGB Hyperparameter Sensitivity Summary

config_name	brier_mean	calibration_slope_mean	roc_auc_mean	pr_auc_mean
baseline_tuned	0.206851	1.042969	0.716004	0.635046
learning_rate_minus20	0.206860	1.049058	0.716107	0.635405
learning_rate_plus20	0.206849	1.043838	0.715896	0.634873
max_depth_minus1	0.206786	1.060979	0.716234	0.635964
max_depth_plus1	0.206791	1.048443	0.715919	0.634774
min_samples_leaf_minus20	0.206790	1.039362	0.715948	0.634640
min_samples_leaf_plus20	0.206990	1.040899	0.715773	0.632685

Table 36. Week 5 Baseline-Family Calibration Comparison

Model family	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
Logistic baseline	0.649822	0.547476	0.231985	0.987174	-0.363811
HGB baseline	0.650200	0.537872	0.225224	1.071831	0.036123

Table 37. Archived Baseline Logistic Primary Metrics

Stream	n_test	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
Uncalibrated	3973	0.649822	0.547476	0.231985	0.987174	-0.363811
Calibrated label stream	3973	0.649822	0.547476	0.231985	0.987174	-0.363811

Table 38. Baseline-family held-out calibration comparison from the earlier logistic-versus-HGB reference stage.

Model	ROC AUC	PR AUC	Brier	Calibration slope	Calibration intercept
logreg	0.65	0.547	0.232	0.987	-0.364
hgb	0.65	0.538	0.225	1.072	0.036

Table 39. Tuned histogram gradient boosting parameter set used as the full-model reference before calibration selection.

Parameter	Value
early_stopping	True
l2_regularization	0.01
learning_rate	0.01
max_depth	5
max_iter	600
max_leaf_nodes	15
min_samples_leaf	120
random_state	2026

Table 40. Compact hyperparameter sensitivity summary for the tuned HGB family.

Configuration	Mean ROC AUC	Mean PR AUC	Mean Brier	Mean slope
baseline_tuned	0.716	0.635	0.2069	1.043
learning_rate_minus20	0.7161	0.6354	0.2069	1.0491
learning_rate_plus20	0.7159	0.6349	0.2068	1.0438
max_depth_minus1	0.7162	0.636	0.2068	1.061
max_depth_plus1	0.7159	0.6348	0.2068	1.0484
min_samples_leaf_minus20	0.7159	0.6346	0.2068	1.0394
min_samples_leaf_plus20	0.7158	0.6327	0.207	1.0409

A.5 Stability, Subgroup, Baseline-Reference, and Error-Slice Diagnostics

The thesis does not treat robustness, subgroup review, baseline reference behavior, and error slicing as interchangeable checks. They answer different questions. Seed perturbations and bootstrap intervals address local numerical stability. Subgroup and slice-level tables address where model behavior is or is not well-documented. The baseline logistic archive clarifies what the simpler reference model was doing before the final calibrated HGB selection.

Placed together in one integrated section, these tables show the full inferential boundary of the project. They do not manufacture certainty. They make uncertainty and heterogeneity inspectable. That is exactly the standard a serious bounded prediction thesis should aim for.

Table 41. Multiseed Stability Summary

seed	roc_auc_mean	pr_auc_mean	brier_mean	slope_mean
2026	0.716004	0.635046	0.206851	1.042969
2027	0.716817	0.636567	0.206756	1.049466
2028	0.715906	0.633901	0.206818	1.039141
2029	0.716441	0.635584	0.206746	1.047280

Table 42. Held-Out Bootstrap Interval Summary

metric	mean	lower_95	upper_95	n_boot
roc_auc	0.716057	0.698754	0.733358	1000
brier	0.206178	0.201062	0.211181	1000
calibration_slope	1.046301	0.956962	1.146544	1000

Table 43. Subgroup Performance Output

subgroup_var	subgroup_value	n	roc_auc	brier	calibration_slope	low_slope_flag	high_error_flag
q2	cat_1	1933	0.681346	0.223363	1.172718	False	False
q2	cat_2	2011	0.644818	0.189929	0.977626	False	False
raceeth	'<NA>'	71	0.680707	0.200654	0.871935	False	False
raceeth	cat_1	270	0.768300	0.200064	1.431038	False	False
raceeth	cat_2	203	0.730654	0.175579	1.251005	False	False
raceeth	cat_3	331	0.742180	0.190177	1.434596	False	False
raceeth	cat_5	1929	0.702611	0.209186	0.968219	False	False
raceeth	cat_6	243	0.711002	0.196585	1.254700	False	False

subgroup_var	subgroup_value	n	roc_auc	brier	calibration_slope	low_slope_flag	high_error_flag
raceeth	cat_7	545	0.677271	0.222562	0.876622	False	False

Table 44. Permutation-Importance Stability Summary

feature_name	mean_importance_neg_brier_score	sd_importance_neg_brier_score	mean_rank_neg_brier_score	fold_consistency_flag
q2	0.022014	0.001783	1.0	True
x_qn25	0.014915	0.001464	2.0	True
x_qn24	0.013076	0.000991	3.0	True
raceeth	0.003486	0.000445	4.0	True
q1	0.000501	0.000388	5.2	True
q3	0.000197	0.000219	5.8	False

Table 45. Review Adequacy and Diagnostic Thresholds

Threshold or diagnostic setting	Value	Role in the thesis
Minimum subgroup size	200	Prevents overreading of thin subgroup cells
Minimum subgroup positives	20	Supports stable slope estimation
Minimum subgroup negatives	20	Supports stable slope estimation
Probability bins	10	Supports appendix-level calibration-bin diagnostics
Flagging logic	`low_slope_flag`, `high_error_flag`	Keeps subgroup review conservative and well-documented

Table 46. Final HGB model error slices by age-proxy strata.

Group	n	Event rate	Error rate @0.5	Mean predicted p	Adequate	Adequacy note
	14	0.286			No	n<200;pos<20;neg<20
cat_1	12	0.583			No	n<200;pos<20;neg<20
cat_2	8	0.0			No	n<200;pos<20;neg<20
cat_3	475	0.381	0.352	0.378	Yes	
cat_4	1158	0.392	0.361	0.4	Yes	

Group	n	Event rate	Error rate @0.5	Mean predicted p	Adequate	Adequacy note
cat_5	1034	0.409	0.372	0.406	Yes	
cat_6	856	0.417	0.377	0.423	Yes	
cat_7	416	0.471	0.37	0.419	Yes	

Table 47. Final HGB model error slices by race-or-ethnicity strata.

Group	n	Event rate	Error rate @0.5	Mean predicted p	Adequate	Adequacy note
	71	0.324			No	n<200
cat_1	270	0.504	0.337	0.474	Yes	
cat_2	203	0.281	0.281	0.345	Yes	
cat_3	331	0.363	0.372	0.396	Yes	
cat_4	17	0.235			No	n<200;pos<20;neg<20
cat_5	1929	0.404	0.39	0.386	Yes	
cat_6	243	0.37	0.342	0.41	Yes	
cat_7	545	0.462	0.363	0.448	Yes	
cat_8	364	0.442	0.346	0.444	Yes	

Table 48. Logistic baseline reference error slices by age-proxy strata.

Group	n	Event rate	Error rate @0.5	Mean predicted p	Adequate	Adequacy note
	14	0.286			No	n<200;pos<20;neg<20
cat_1	12	0.583			No	n<200;pos<20;neg<20
cat_2	8	0.0			No	n<200;pos<20;neg<20
cat_3	475	0.381	0.366	0.462	Yes	
cat_4	1158	0.392	0.37	0.482	Yes	
cat_5	1034	0.409	0.386	0.493	Yes	
cat_6	856	0.417	0.387	0.515	Yes	
cat_7	416	0.471	0.373	0.503	Yes	

Table 49. Logistic baseline reference error slices by race-or-ethnicity strata.

Group	n	Event rate	Error rate @0.5	Mean predicted p	Adequate	Adequacy note
	71	0.324			No	n<200
cat_1	270	0.504	0.337	0.57	Yes	
cat_2	203	0.281	0.384	0.41	Yes	
cat_3	331	0.363	0.35	0.484	Yes	
cat_4	17	0.235			No	n<200;pos<20;neg<20
cat_5	1929	0.404	0.402	0.467	Yes	
cat_6	243	0.37	0.337	0.492	Yes	
cat_7	545	0.462	0.363	0.549	Yes	
cat_8	364	0.442	0.341	0.539	Yes	

Table 50. Logistic baseline coefficients and odds ratios from the fixed seed-2026 reference fit.

Variable	Level	Coefficient	Odds ratio
q1	cat_1	0.617	1.854
q1	cat_2	-0.574	0.563
q1	cat_3	-0.162	0.851
q1	cat_4	-0.026	0.974
q1	cat_5	0.094	1.099
q1	cat_6	0.195	1.215
q1	cat_7	0.203	1.226
q2	cat_2	-1.05	0.35
q3	cat_1	0.062	1.064
q3	cat_2	-0.055	0.946
q3	cat_3	-0.03	0.97
q3	cat_4	-0.116	0.89
q3	cat_5	0.488	1.628
raceeth	cat_1	0.337	1.401
raceeth	cat_2	-0.197	0.821

Variable	Level	Coefficient	Odds ratio
raceeth	cat_3	0.009	1.009
raceeth	cat_4	-0.353	0.703
raceeth	cat_5	-0.07	0.932
raceeth	cat_6	0.081	1.085
raceeth	cat_7	0.24	1.271
raceeth	cat_8	0.301	1.351

A.6 Diagnostic Section Summary

This integrated diagnostic section expands the manuscript without changing the scientific claim. The bullying block still improves held-out prediction beyond the narrow demographic baseline, and the final HGB plus Platt model still provides the strongest balance of discrimination and probability quality.

What changes after bringing these diagnostics into the body is the density of visible evidence. The reader can inspect the coding architecture, prevalence structure, metric definitions, candidate neighborhood, stability summaries, and subgroup error slices directly in the thesis itself. That makes the argument easier to evaluate and less dependent on unseen supporting material.

Appendix B. Robustness and Threshold-Oriented Diagnostics

This appendix preserves the subgroup, threshold-slice, local-sensitivity, and seed-stability diagnostics used to audit how the selected modeling approach behaves beyond the primary held-out comparison. These materials are interpretive and bounding devices rather than alternative selection criteria.

B.1 Why a Threshold- and Error-Focused Section Belongs in the Body

The final model is selected on held-out probability quality rather than on a single classification cut point, yet a reader still needs to know what the archived diagnostic outputs look like when probabilities are translated into concrete errors. That is the reason for this section. The earlier results and integrated diagnostics sections established which candidate wins and why. They did not fully unpack how the archived HGB-versus-logistic reference diagnostics behave once predicted probabilities are viewed through a threshold-based diagnostic frame. A senior thesis that claims probability-first rigor should still show the reader what thresholded error patterns look like, which groups are most affected, and where the probability outputs remain closer to observed event rates than a simpler reference model.

That threshold-based diagnostic frame must be handled carefully. The present project is not a deployment paper, and the 0.5 threshold used in the archived error-slice outputs is not treated here as a recommended decision threshold. It is a reference cut that helps expose comparative behavior. Because the overall event rate for the primary outcome is about 0.41 in the analysis-ready table, a 0.5 cut is already stricter than the base rate. The value of the threshold exercise is therefore not that it gives a policy rule. Its value is that it reveals how the archived HGB baseline reference run behaves relative to the logistic baseline reference once the smoother probability story is translated into concrete mistakes.

A second reason this section belongs in the body is that the thesis makes calibration a co-primary criterion. Calibration claims are easier to trust when the manuscript also shows what the archived probability streams imply for subgroup error structure. If a model appears well calibrated in aggregate but creates erratic or obviously unbalanced subgroup behavior, a reader would rightly question how meaningful the calibration story really is. The body therefore needs a location where that bridge is made explicitly rather than left implicit.

B.2 Age-Proxy Error Structure in the Archived HGB Baseline Reference Run

Among age-proxy strata with adequate sample size, the archived HGB baseline reference run shows consistently lower 0.5-threshold error than the logistic baseline reference. Across strata `cat_3` through `cat_7`, the mean threshold-error difference is 0.010 in favor of HGB. The advantage is not enormous, which is important to say directly. This is not a case where the archived HGB reference run transforms the classification problem. It is a case where the HGB

reference run preserves similar ranking behavior while reducing probability distortion enough that thresholded error also trends downward when the same archived cut point is applied.

The stronger pattern is calibration alignment inside strata. For the adequate q1 groups, the mean absolute gap between the HGB mean predicted probability and the observed event rate is only 0.014, whereas the corresponding gap for the logistic reference is 0.077. That contrast matters more than the raw threshold gain because it tells the reader why the threshold results move in the direction they do. The archived HGB baseline reference run is not merely shuffling cases around the 0.5 line. It is producing subgroup-average probabilities that sit much closer to the empirical rates observed within those same strata.

The remaining imperfections are also visible. The oldest adequate q1 stratum still shows mild underprediction under HGB, and no model in this thesis eliminates subgroup heterogeneity. That is exactly why these results belong in the paper. A calibrated model is not a perfectly calibrated model, and a better probability engine is not a universal solution to subgroup complexity. The proper reading is narrower. Within the locked design, the archived HGB baseline reference run reduces both mean probability misalignment and threshold error across the adequately sized age-proxy strata without erasing the underlying heterogeneity of the survey population.

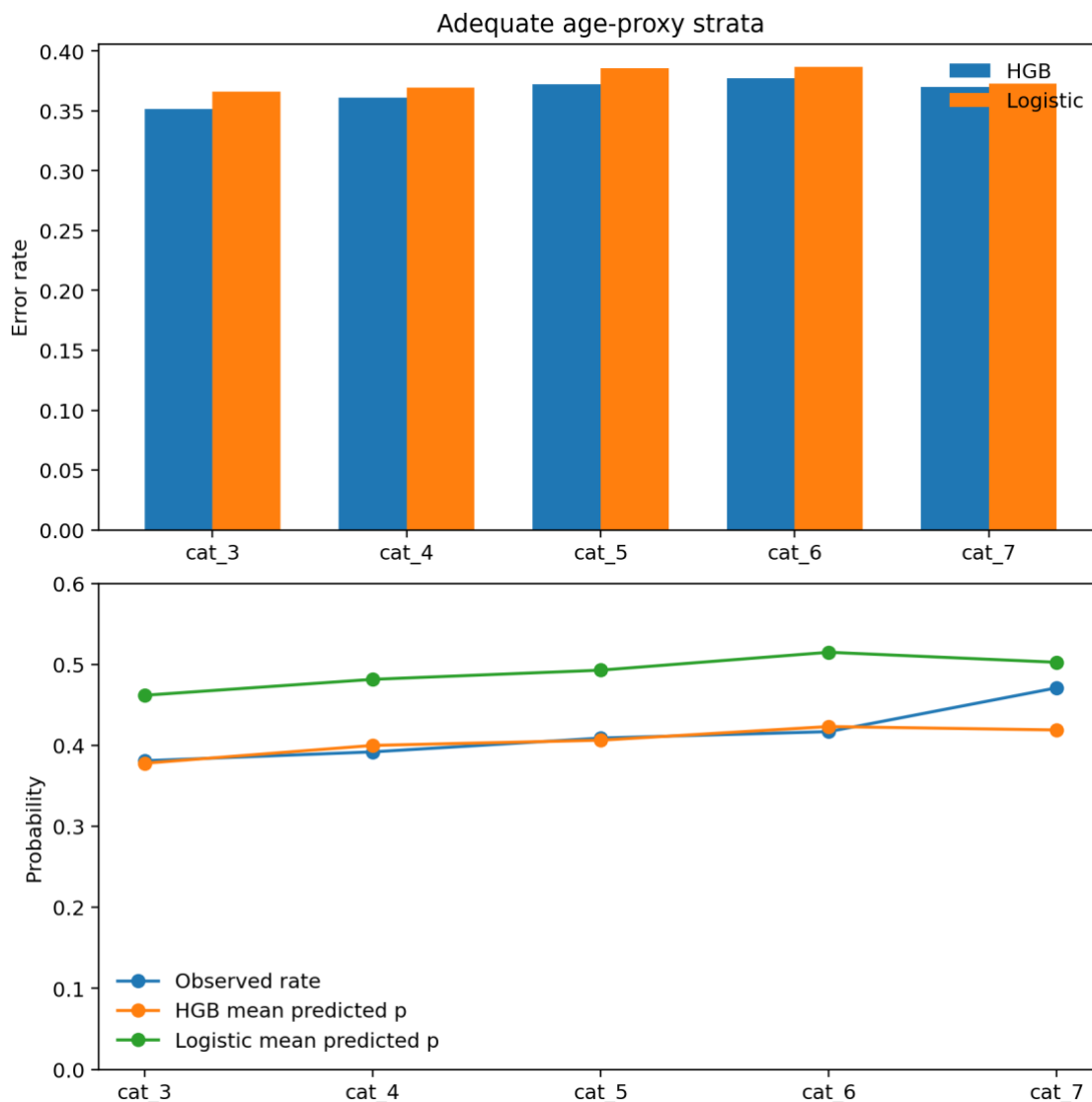
Table 51. Archived HGB baseline-reference error slices across adequate age-proxy strata.

Age proxy	n	Observed rate	Error rate @0.5	Mean predicted p
cat_3	475	0.381	0.352	0.378
cat_4	1158	0.392	0.361	0.400
cat_5	1034	0.409	0.372	0.406
cat_6	856	0.417	0.377	0.423
cat_7	416	0.471	0.370	0.419

Table 52. Logistic reference error slices across adequate age-proxy strata.

Age proxy	n	Observed rate	Error rate @0.5	Mean predicted p
cat_3	475	0.381	0.366	0.462
cat_4	1158	0.392	0.370	0.482
cat_5	1034	0.409	0.386	0.493
cat_6	856	0.417	0.387	0.515
cat_7	416	0.471	0.373	0.503

Figure 11. Age-proxy threshold error and mean predicted probability comparison for the archived HGB baseline reference run versus the logistic baseline reference. The upper panel shows 0.5-threshold error rates. The lower panel shows how closely each model's subgroup-average predicted probability tracks the observed event rate.



B.3 Race-or-Ethnicity-Proxy Error Structure

The race-or-ethnicity-proxy error slices show a slightly more mixed threshold story than the age-proxy slices, but they still favor the archived HGB baseline reference run on average. Across adequate strata, the mean threshold-error difference is 0.012 in favor of HGB. That average hides variation. Some strata show nearly identical threshold error under both models, and a small number show marginally worse HGB error at the archived 0.5 cut. Those cases are analytically useful because they illustrate why threshold error cannot be the only criterion. A model can be better calibrated overall and still fail to improve every subgroup under one reference threshold.

The probability-alignment story is much cleaner. The mean absolute gap between subgroup-average predicted probability and subgroup observed event rate is 0.029 for HGB versus 0.098 for the logistic reference. In other words, the archived HGB baseline reference run compresses the chronic overprediction that appears in the logistic subgroup means. That overprediction is visible in nearly every adequate race/ethnic stratum. The HGB reference run does not remove every residual, but it pulls the subgroup-average probabilities much closer to the corresponding observed rates.

The most instructive case is the lower-rate race/ethnic cat_2 stratum. Under the logistic reference, the subgroup mean predicted probability sits far above the observed event rate. Under the archived HGB baseline reference run, the same subgroup remains somewhat overpredicted, but the gap narrows materially and threshold error falls sharply. Other strata are more balanced. In cat_1 and cat_7, threshold error is identical across models, but the archived HGB reference run still produces subgroup-average probabilities that are much closer to the observed rates. That is exactly the type of result the thesis is designed to value. Even when one threshold does not produce a visible error improvement, probability quality can still improve in a way that matters for interpretation.

These patterns also reinforce the subgroup-audit logic introduced earlier. The race/ethnic slices are informative because they are large enough to compute and diverse enough to reveal where aggregate calibration can conceal real within-group differences. They are not a full fairness analysis, and the thesis does not present them as one. They are a validity audit for a sensitive prediction task on survey data.

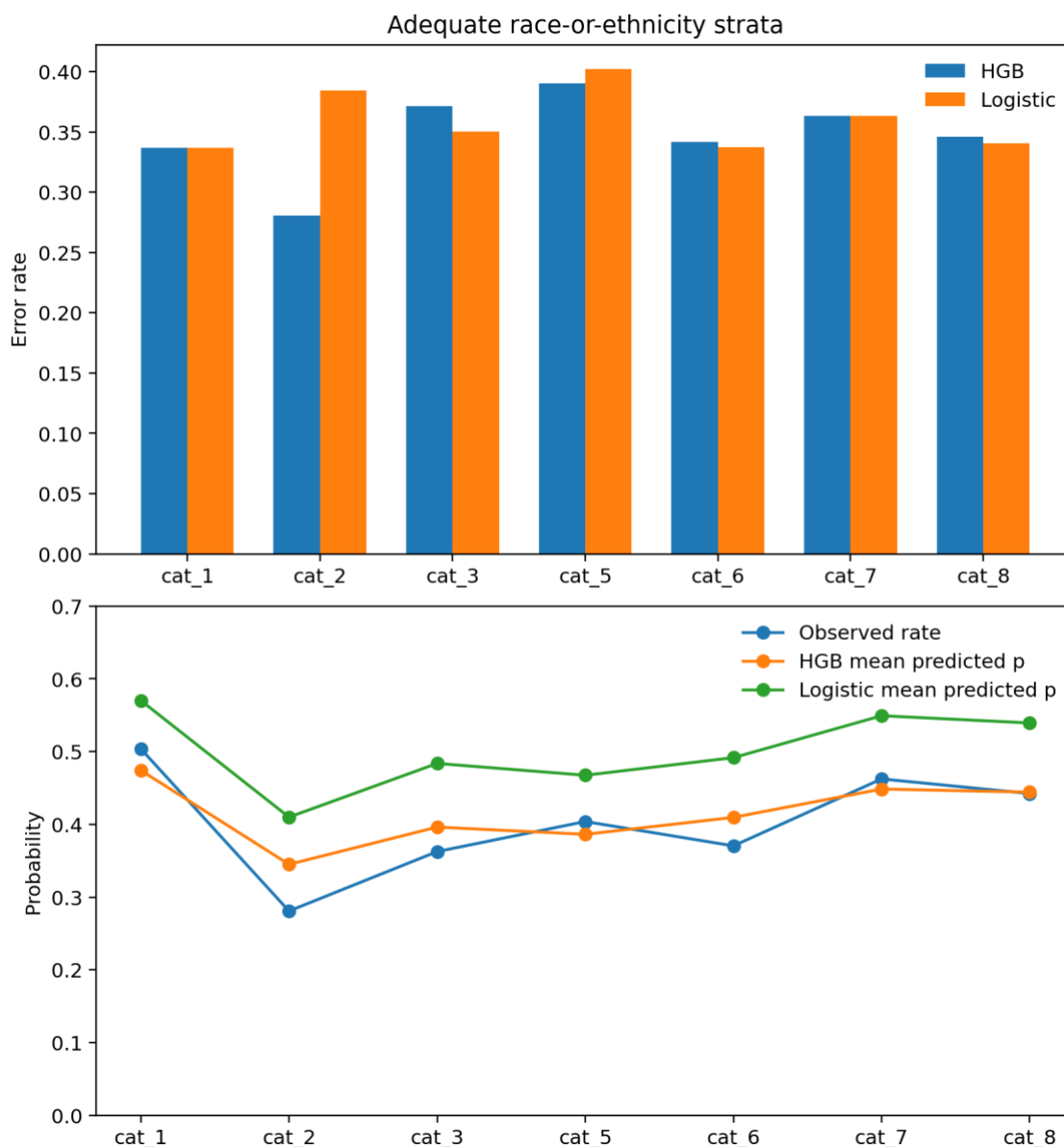
Table 53. Archived HGB baseline-reference error slices across adequate race-or-ethnicity-proxy strata.

Race/eth proxy	n	Observed rate	Error rate @0.5	Mean predicted p
cat_1	270	0.504	0.337	0.474
cat_2	203	0.281	0.281	0.345
cat_3	331	0.363	0.372	0.396
cat_5	1929	0.404	0.390	0.386
cat_6	243	0.370	0.342	0.410
cat_7	545	0.462	0.363	0.448
cat_8	364	0.442	0.346	0.444

Table 54. Logistic reference error slices across adequate race-or-ethnicity-proxy strata.

Race/eth proxy	n	Observed rate	Error rate @0.5	Mean predicted p
cat_1	270	0.504	0.337	0.570
cat_2	203	0.281	0.384	0.410
cat_3	331	0.363	0.350	0.484
cat_5	1929	0.404	0.402	0.467
cat_6	243	0.370	0.337	0.492
cat_7	545	0.462	0.363	0.549
cat_8	364	0.442	0.341	0.539

Figure 12. Race-or-ethnicity-proxy threshold error and mean predicted probability comparison for the archived HGB baseline reference run versus the logistic baseline reference. The figure highlights the stronger compression of subgroup overprediction under HGB.



B.4 Why Threshold Error and Calibration Are Not the Same Quantity

The q1 and raceeth slices make a broader methodological point. Threshold error and calibration are related, but they are not interchangeable. Threshold error depends on a chosen cut point and on where the score distribution sits relative to that cut. Calibration is about the relationship between predicted probability and observed frequency across the score range. A model can improve calibration while leaving threshold error unchanged for some subgroups. It can also

worsen threshold error at one cut point while still generating probabilities that are more faithful to subgroup event rates overall.

That distinction matters because the thesis does not claim that the selected final model should be used to trigger action at probability 0.5. The archived threshold is only a diagnostic. The actual argument of the manuscript remains probability-first. Better subgroup-average alignment in the archived HGB reference diagnostics, lower held-out Brier score, and slope closer to one under Platt calibration are all more central to the final claim than a handful of threshold-specific subgroup comparisons. At the same time, a body-only thesis would feel incomplete if it never showed the reader what those threshold-specific comparisons look like. This section therefore clarifies the relationship rather than pretending that one type of evidence can substitute for the other.

B.5 Subgroup Performance Diagnostics Beyond Error Rate

The subgroup performance output adds a second diagnostic layer beyond the archived 0.5-threshold slices. For q2 and raceeth, the archived uncalibrated full HGB reference run retains defined ROC AUC and calibration slope in the adequately sized subgroups, and none of the defined-slope groups falls below the thesis low-slope flag of 0.8. This is not a claim of subgroup parity. It is a claim that the most concerning failure mode specified in the governance framework was not observed in the available subgroup output.

Variation remains substantial. Brier values are lower in some strata than in others, and subgroup slopes range from values close to one to values well above one. That is why the thesis treats subgroup review as an audit trigger rather than as a decorative fairness section. The correct interpretation is that the selected model stays inside the project's pre-specified guardrails while still displaying enough heterogeneity to justify caution about any use beyond research-stage analysis.

Table 55. Subgroup performance diagnostics for the archived uncalibrated full HGB reference run. Defined metrics are reported only where archived adequacy conditions are met.

Subgroup variable	Subgroup value	n	ROC AUC	Brier	Slope	ROC defined	Slope defined	Low slope flag	High error flag
q2	NA	29	NA	0.188	NA	False	False	False	False
q2	cat_1	1933	0.681	0.223	1.173	True	True	False	False
q2	cat_2	2011	0.645	0.190	0.978	True	True	False	False
raceeth	NA	71	0.681	0.201	0.872	True	True	False	False
raceeth	cat_1	270	0.768	0.200	1.431	True	True	False	False
raceeth	cat_2	203	0.731	0.176	1.251	True	True	False	False

Subgroup variable	Subgroup value	n	ROC AUC	Brier	Slope	ROC defined	Slope defined	Low slope flag	High error flag
raceeth	cat_3	331	0.742	0.190	1.435	True	True	False	False
raceeth	cat_4	17	NA	0.171	NA	False	False	False	False
raceeth	cat_5	1929	0.703	0.209	0.968	True	True	False	False
raceeth	cat_6	243	0.711	0.197	1.255	True	True	False	False
raceeth	cat_7	545	0.677	0.223	0.877	True	True	False	False
raceeth	cat_8	364	0.728	0.211	0.997	True	True	False	False

B.6 Local Stability Around the Tuned HGB Configuration

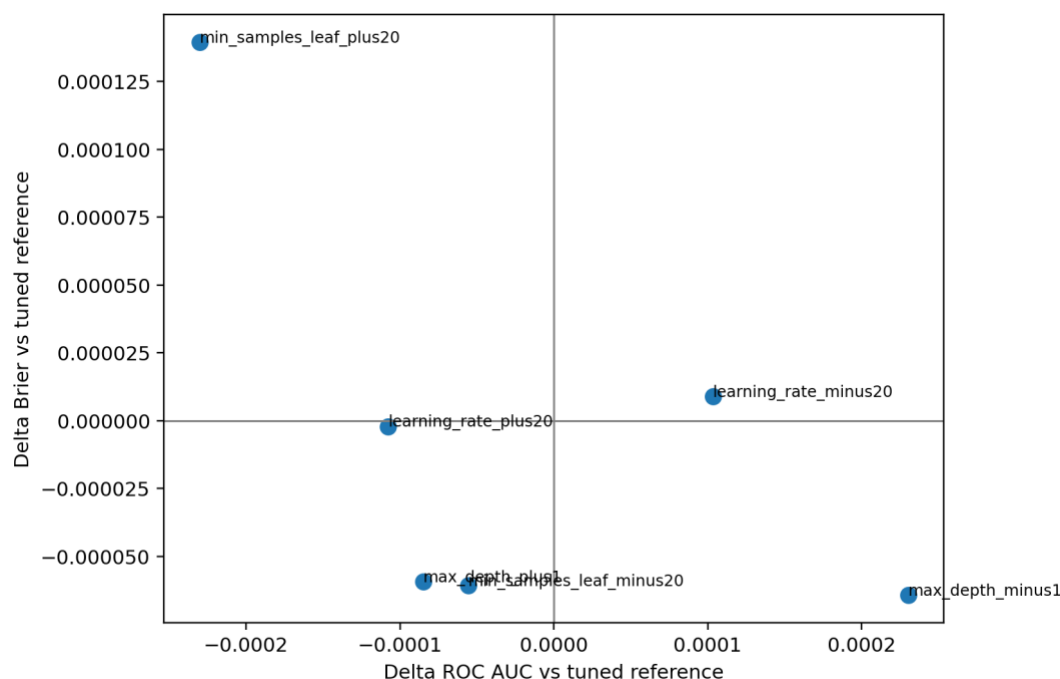
Interpretation is stronger when the nearby tuned full HGB configuration is not fragile. The one-factor hyperparameter sensitivity analysis supports that stability claim. Across six local perturbations around the tuned HGB point, the maximum absolute change in cross-validation Brier is only 0.000139. The ROC AUC and PR AUC changes are similarly small. In practical terms, the archived tuned configuration behaves like a stable neighborhood rather than a single lucky coordinate.

That does not mean tuning no longer matters. The tuned point still anchors the archived full HGB reference family, and some perturbations shift slope or intercept more than others. The key point is that the HGB family does not appear to look strong only because of an isolated hyperparameter accident. This matters for interpretation because brittle model choices are harder to defend in an undergraduate thesis with a narrow feature scope and a frozen protocol. A stable local neighborhood gives the archived reference family more credibility as a substantive empirical object rather than a numerical curiosity.

Table 56. Local hyperparameter sensitivity around the tuned HGB reference configuration.

Configuration	Brier	ROC AUC	PR AUC	Slope	Intercept
baseline_tuned	0.207	0.716	0.635	1.043	0.015
learning_rate_minus20	0.207	0.716	0.635	1.049	0.017
learning_rate_plus20	0.207	0.716	0.635	1.044	0.015
max_depth_minus1	0.207	0.716	0.636	1.061	0.022
max_depth_plus1	0.207	0.716	0.635	1.048	0.017
min_samples_leaf_minus20	0.207	0.716	0.635	1.039	0.014
min_samples_leaf_plus20	0.207	0.716	0.633	1.041	0.014

Figure 13. Local HGB hyperparameter sensitivity around the archived tuned reference configuration. Each point shows the change in cross-validation ROC AUC and Brier relative to that reference point.



B.7 Cross-Seed Consistency and Bootstrap Uncertainty

The multiseed summary provides another stability check. Across seeds 2026 to 2029, the range of cross-validation ROC AUC means is 0.000911, the range of Brier means is 0.000105, and the range of slope means is 0.010325. Those ranges are very small in absolute terms. The archived uncalibrated full HGB reference configuration is therefore not being held together by one favorable random partition of the training data. The frozen holdout remains fixed, and the internal training-fold realizations still tell a highly consistent story.

The held-out bootstrap intervals complete the picture. They show that the archived reference run's held-out ROC AUC, Brier, and calibration slope each lie inside reasonably narrow uncertainty bands under stratified resampling. The bootstrap does not certify transportability. It does, however, show that the paper's family-level stability story is not hanging on a single unstable estimate. For a senior thesis, that is the right level of claim discipline. The uncertainty intervals are used to frame precision, not to overstate certainty.

Table 57. Cross-seed stability summary for the archived uncalibrated full HGB reference configuration.

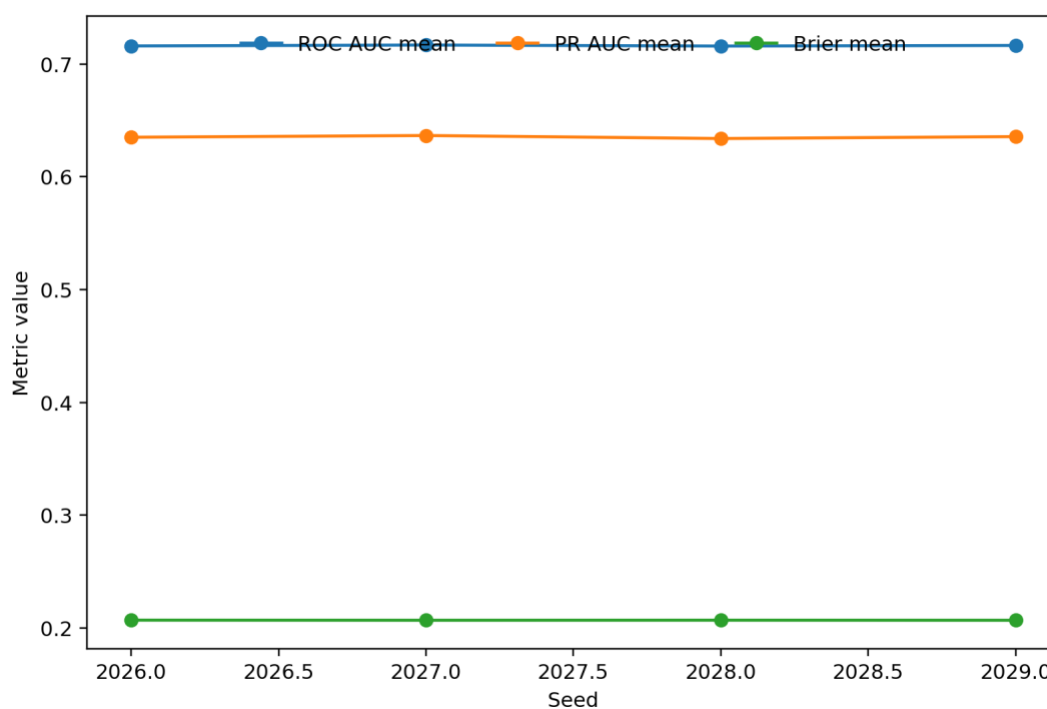
Seed	ROC AUC mean	PR AUC mean	Brier mean	Slope mean
2026.0	0.716	0.635	0.207	1.043
2027.0	0.717	0.637	0.207	1.049

Seed	ROC AUC mean	PR AUC mean	Brier mean	Slope mean
2028.0	0.716	0.634	0.207	1.039
2029.0	0.716	0.636	0.207	1.047

Table 58. Held-out bootstrap interval summary for the archived uncalibrated full HGB reference configuration.

Metric	Mean	Lower 95%	Upper 95%	Bootstrap draws
roc_auc	0.716	0.699	0.733	1000
brier	0.206	0.201	0.211	1000
calibration_slope	1.046	0.957	1.147	1000

Figure 14. Cross-seed stability of the archived uncalibrated full HGB reference configuration across the multiseed run set. The tight clustering reinforces that the family-level results are not driven by one internal fold realization.



B.8 Threshold-Oriented Interpretation Without Deployment Claims

The section title uses threshold-oriented in a narrow sense. It refers to what the archived diagnostics show when probabilities are translated into concrete errors, subgroup averages, or review flags. It does not imply that the thesis recommends deployment. In fact, the opposite is true. The more closely the manuscript inspects threshold behavior, subgroup differences, and stability conditions, the clearer it becomes that any deployment-adjacent use would require

stricter recalibration, refreshed subgroup review, and an external validation setting that this thesis does not provide.

That distinction is one of the strengths of the paper. A thesis can examine decision-facing behavior without pretending that its own evidence is sufficient for real-world decision support. The current section therefore strengthens the governance posture of the thesis. It shows what the selected model does under an archived threshold, where it improves on the logistic reference, and where caution is still warranted. The next section turns that caution into an explicit main-text governance framework.

B.9 Section Summary

Three points emerge from this section. First, the archived HGB baseline reference run generally reduces threshold error relative to the logistic reference in the adequately sized q1 and raceeth slices, although the gains are modest and not universal. Second, the stronger and more stable advantage lies in subgroup probability alignment. The archived HGB baseline reference run compresses the systematic subgroup overprediction seen in the logistic reference. Third, the local stability, multiseed, and bootstrap evidence shows that the archived full HGB reference configuration remains coherent when examined from several different diagnostic angles.

Taken together, these results justify a richer interpretation of the earlier probability-first findings. The selected final model is still the best candidate under the lexicographic held-out rule. The archived baseline-reference diagnostics and full-HGB robustness outputs then add bounded context about subgroup errors, subgroup-average probabilities, and stability under nearby perturbations. That does not convert the project into a deployment study. It does make the thesis body more complete.